VACCINES

Stabilization of the H5 clade 2.3.4.4b hemagglutinin improves vaccine-elicited neutralizing antibody responses in mice

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Transmission of highly pathogenic avian influenza from H5 clade 2.3.4.4b has expanded in recent years to infect large populations of birds and mammals, heightening the risk of a human pandemic. Influenza viruses that are adapted to transmission in birds and a variety of mammals tend to have a less stable hemagglutinin (HA) than seasonal influenza viruses, enabling membrane fusion at comparatively higher pH levels. Here, we combined five mutations in the H5 HA that increased its melting temperature and promoted stable closure of the HA trimer. Structural analysis by cryo–electron microscopy revealed that the stabilizing mutations create several new hydrophobic interactions while maintaining the local HA structure. We found that vaccinating mice with stabilized H5 HA immunogens resulted in higher hemagglutination inhibition and neutralization titers than nonstabilized comparators. Epitope mapping of vaccine-elicited polyclonal antibody responses using negative-stain electron microscopy and deep mutational scanning showed that site E on the side of the HA receptor binding domain was immunodominant across all groups; however, the stabilized immunogens shifted responses toward the receptor binding site, which elicited a higher proportion of neutralizing antibodies. Consistent with these findings, stabilized H5 HA immunogens delivered as messenger RNA–lipid nanoparticle (mRNA-LNP) vaccines protected mice against H5N1 challenge. These findings highlight that H5 HA–stabilizing mutations enhance the quality of antibody responses across different vaccine formats, underscoring their potential to improve pandemic preparedness vaccines targeting viruses from this widely circulating clade.

INTRODUCTION

Since 1996, highly pathogenic avian influenza (HPAI) H5N1 viruses have been spreading at a large scale among both wild and domesticated birds, with spillover infections occurring in several mammalian species, including humans (*1*). By 2021, clade 2.3.4.4b had emerged as the dominant global strain and was introduced to North America through wild bird migration (*2*, *3*). In 2024, large H5N1 outbreaks in both dairy cattle and domesticated poultry across the United States devastated both industries and led to several dozen infections in humans (*4*–*7*). Large-scale infections of HPAI H5N1 have been observed in several other mammals as well, including sea lions, foxes, llamas, and cats, underscoring its adaptability (*8*). Underlying these outbreaks, recent studies suggest that H5N1 transmission continues to increase through multiple evolutionary mechanisms (*9*, *10*). This is especially concerning given in vitro evidence that only a single mutation in an avian H5 hemagglutinin (HA) is needed to bind to a human receptor (*11*, *12*). The persistent evolution and broad circulation of HPAI H5N1 across multiple animal species pose an ongoing threat of a potential human pandemic.

Given the high evolutionary rate of HPAI H5N1, a quick turnaround time to produce new vaccines is necessary. Greatly improved production speed has been achieved through the advent of mRNA vaccines, which enable a single manufacturing process for any sequence (*13*, *14*). Several previous studies using membrane-anchored H5 HA mRNA vaccines have shown elicitation of neutralizing antibodies in mice and ferrets, and human clinical trials have been initiated (*15*–*20*). Recent work from our lab combining mRNA vaccines with the enhanced immunogenicity of multivalent antigen display showed that an mRNA-launched computationally designed protein nanoparticle displaying 60 copies of the severe acute respiratory syndrome coronavirus 2 spike receptor binding domain (RBD) elicited higher neutralization titers in mice compared with membrane-anchored spike (*21*). These findings highlight the potential of integrating mRNA technology with protein nanoparticle display to swiftly generate effective vaccines against rapidly evolving viruses like HPAI H5.

Despite the continued focus on HA as the primary influenza vaccine antigen, and unlike many other class I viral fusion proteins (*22*–*27*), there has been little work examining whether increasing HA protein stability improves immunogenicity. Only recently, Milder *et al.* (*28*) showed that the poor expression and monomeric nature of several HA subtypes when expressed without trimerization domains, including H5, can be improved through structure-based vaccine design. Even with the addition of foldon or the trimeric nanoparticle component I53_dn5B as trimerization domains, considerable instability and opening of HA at its trimer interface were observed. To improve HA stability, they introduced hydrophobic mutations in two pH-switch regions conserved across all major influenza A subtypes that markedly improved expression and trimerization. In a separate study, stability-enhancing mutations in H5 HA were identified through successive heat treatments of a virus library incorporating random mutations (*29*). Further evidence of the marked instability

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of H5 HA is seen in the increased pH of fusion of this subtype as compared with seasonal influenza strains (*30*). Perhaps relatedly, an inactivated H5 vaccine has been observed to be less immunogenic as compared with the seasonal inactivated vaccine in ferrets (*31*). Despite this evidence for H5 protein instability, as well as the previous identification of stabilizing mutations, the impact of stabilization on immunogenicity in vivo remains incompletely understood. Here, we introduced mutations into the clade 2.3.4.4b influenza H5 HA protein to improve its stability. We then compared the wild-type and stabilized HA in several protein and mRNA vaccine formats in preclinical models. We found that delivering stabilized protein nanoparticle and membrane-anchored H5 HAs as mRNA–lipid nanoparticles (LNPs) elicited a higher proportion of receptor binding site (RBS)–directed responses and potent HA inhibition (HAI) and neutralizing activity, thereby imparting protection against H5N1 challenge in mice.

RESULTS

Stabilizing mutations increase the melting temperature and trimeric closure of H5 HA

To stabilize the prefusion conformation of HPAI A/Texas/37/2024 (TX24) H5 HA, combinations of previously identified mutations from multiple studies were tested for increased thermal stability using nano–differential scanning fluorimetry (nanoDSF; fig. S1A) (*11*, *28*, *29*, *32*). The H5 HA variant with the highest melting temperature (T_m) incorporated the following mutations in the HA2 domain: H355F/K380M/R397M/L418I/E432L [based on mature H3 HA numbering (*33*)] (Fig. 1A). Hereafter, we refer to this stabilized HA as H5-FMLMI, because it contains mutations to these amino acid identities at stabilized sites. This H5 HA was genetically fused via flexible $(GS)_n$ linkers to a foldon trimerization domain or the previously described one-component de novo nanoparticle RC_I_1 (Fig. 1B and table S1) (*34*). In addition, all H5 HAs used in this study contained the Y98F mutation to facilitate production (*35*, *36*) and have a mutated polybasic cleavage site, resulting in HA0 constructs. The T_m values of H5-FMLMI-foldon were 77.8° and 76.4°C, whereas the T_m values of wild-type H5-foldon were 55.6° and 60.0°C, at pH 8.0 and pH 5.5, respectively (Fig. 1C). The T_m values of H5-FMLMI-RC_I_1 were 78.5° and 74.3°C, whereas the T_m values of H5-RC_I_1 were 52.4° and 36.7°C, at pH 8.0 and pH 5.5, respectively. The low T_m for H5-RC_I_1 at pH 5.5 indicated that this scaffold is relatively unstable at low pH. However, this instability was mitigated by the stabilizing mutations in the HA antigen, which resulted in similar T_m values at pH 5.5 for H5-FMLMI-foldon and H5-FMLMI-RC_I_1.

Final stabilized and nonstabilized H5-foldon and H5-RC_I_1 nanoparticle constructs were scaled up for production in human embryonic kidney (HEK) 293F cells. After immobilized metal affinity chromatography, proteins were further purified by size exclusion chromatography (SEC), resulting in chromatograms with single elution peaks at expected retention volumes for all four constructs (Fig. 1D). Final protein yields per liter of cell culture were 12.8 mg/liter for H5-foldon, 26.8 mg/liter for H5-FMLMI-foldon, 10.4 mg/liter for H5-RC_I_1, and 4.4 mg/liter for H5-FMLMI-RC_I_1. SDS–polyacrylamide gel electrophoresis of these elution peaks showed a single band that migrated at the expected molecular weight for all constructs (fig. S1B). Negative-stain electron microscopy (nsEM) micrographs and two-dimensional (2D) class averages showed that both H5 foldon constructs formed trimeric HAs and both H5 RC_I_1 constructs formed intact, monodisperse nanoparticles (fig. S1C). Dynamic light scattering (DLS) showed the expected size for both H5 foldon constructs, but both H5 RC_I_1 nanoparticles had a hydrodynamic radius of 30 nm, 10 nm smaller than the H5 RC_I_1 design model (fig. S1D) (*34*). The mass calculated using mass photometry (MP) of H5-FMLMI-RC_I_1 was consistent with approximately eight trimers per nanoparticle (fig. S1E). nsEM, DLS, and MP together suggest that both H5 RC_I_1 constructs formed octahedra rather than the intended icosahedra.

Biolayer interferometry (BLI) against the anti-RBS monoclonal antibody (mAb) 12D03 (*37*), anti–lateral patch mAb 7D11 (*37*), and

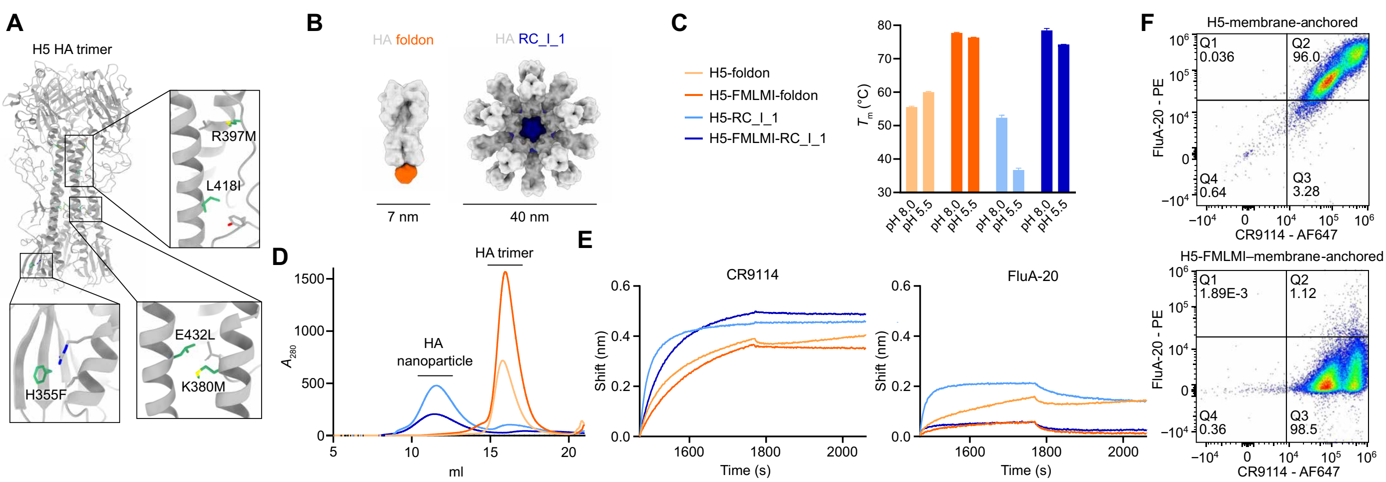

Fig. 1. Stabilizing mutations in H5 increase the thermostability and reduce binding of internally directed mAbs. (**A**) Stabilizing mutations H355F/K380M/R397M/L418I/E432L (green) highlighted in the A/Texas/37/2024 crystal structure (PDB ID: 9DIQ). Key interacting side chains are shown in gray. (**B**) Surface models of HA foldon trimer (foldon in orange) and HA RC_I_1 one-component nanoparticle (nanoparticle in blue). (**C**) T_m values of H5 HA constructs at pH 8.0 and pH 5.5 as measured by nanoDSF. The bars and error bars represent the means ± SEM for technical replicates ($n = 3$). (**D**) SEC chromatograms of H5 HA constructs. Same legend as (C). A_{280}, absorbance at 280 nm. (**E**) BLI of H5 HA constructs against anti-stem mAb CR9114 (left) and anti–trimer interface mAb FluA-20 (right). Same legend as (C). (**F**) Flow cytometry plots of CR9114 and FluA-20 binding to membrane-anchored wild-type (upper) and stabilized (lower) H5 HA constructs expressed on the surface of HEK293F cells.

anti-stem mAb CR9114 (*38*) all showed high binding for all four H5 HA constructs (Fig. 1E and fig. S1F). FluA-20 is a mAb that binds to the interface between HA monomers in the trimer, so it is only able to bind to partially open HA trimers (*39*). High binding was observed for FluA-20 against both H5-foldon and H5-RC_I_1 compared with essentially no FluA-20 binding in their stabilized counterparts, suggesting that these stabilizing mutations induce the closure of the trimer interface in these constructs (Fig. 1E). Conformation-specific mAbs were also used to probe the antigenicity of membrane-anchored H5 HA on the surface of HEK293F cells (Fig. 1F). High anti-stem mAb CR9114 and FluA-20 binding was observed for the H5 membrane-anchored construct, whereas high CR9114 and minimal FluA-20 binding was observed for the H5-FMLMI–membrane-anchored construct. These results indicate that these H5 membrane-anchored constructs express similarly but that the head domains of the wild-type H5 HA are open or monomeric, whereas the FMLMI-stabilizing mutations encourage closed trimer formation.

Cryo–electron microscopy reconstruction of H5-FMLMI-foldon reveals hydrophobic interactions formed by stabilizing mutations

To understand how the FMLMI mutations contributed to H5 HA stabilization and its conformation, we determined a 3.2-Å-resolution reconstruction of H5-FMLMI-foldon by cryo–electron microscopy (cryo-EM) (Fig. 2A; fig. S2, A to D; and table S2). A preferred orientation was observed for this sample (fig. S2B), so data were collected at tilt angles of 0° and 30°, a method previously shown to improve this frequently encountered issue in HA trimer cryo-EM reconstructions (*40*). The backbone root mean square deviation (RMSD) to a TX24 H5 HA cryo-EM structure [Protein Data Bank (PBD) ID: 9DWE] was 3.5 Å over the entire trimer (Fig. 2B) (*10*). Several factors could contribute to this deviation, including the introduction of stabilizing mutations, the presence of a bound antibody in the wild-type H5 HA structure, the flexibility of the base of HA also observed in other H5 cryo-EM structures (*37*), or structural artifacts because of the preferred orientation. Despite this relatively high RMSD over the entire trimer, the RMSD was low when aligning on individual epitopes, with an RMSD of 1 Å for the stem epitope and an RMSD of 0.6 Å for the RBS epitope (Fig. 2B). This is consistent with the similar binding we observed between stabilized and unstabilized H5 HA constructs in vitro for anti-RBS and anti-stem mAbs (Fig. 1E and fig. S1F). We concluded that the addition of stabilizing mutations still preserved the local structure of these key HA epitopes. Further analysis of the stabilizing mutations in the cryo-EM reconstruction showed formation of several hydrophobic interactions (Fig. 2, C and D). R397M replaced polar interactions of R397 with N410 and N408 with hydrophobic contacts to L409. Despite the relatively weak density for R397M, modeling of the residues surrounding this position shows that this is the most plausible placement of this side chain (fig. S2E). L418I appeared to encourage hydrophobic packing with Y308. E432L and K380M replaced this internal salt bridge with hydrophobic interactions with L428, I384, and M30. Last, H355F replaced a pH-dependent ionic interaction of H355 to R482 with a hydrophobic contact to M478. Similar substitutions that create new hydrophobic interactions have been previously observed in other HA stabilization efforts (*28*).

H5 HA–stabilizing mutations increase neutralizing antibody responses in mice

Next, we evaluated these stabilized H5 HA constructs as both protein- and nucleoside-modified mRNA-LNP–delivered vaccines in BALB/c mice. Mice were subcutaneously immunized at weeks 0 and 4 with 21.8 pmol of total protein (1.37 μg for H5 foldon constructs and 1.5 μg for H5 nanoparticles) formulated with AddaVax or with 1 μg of mRNA-LNP (Fig. 3A and fig. S3A). Vaccine-elicited serum antibody responses were assessed 2 weeks after prime and after boost.

To assess serum binding, vaccine-matched TX24 and vaccine-mismatched A/Indonesia/05/2005 (IN05) HAs genetically fused to the trimeric nanoparticle component I53_dn5B (*41*, *42*) were used in enzyme-linked immunosorbent assays (ELISAs). After prime, there were higher TX24 binding titers in the stabilized nanoparticle ($P < 0.0001$ for protein-delivered; $P < 0.01$ for mRNA-LNP–delivered) and membrane-anchored groups ($P < 0.05$) compared with the similar construct without stabilization (Fig. 3, B and C). Among mRNA-LNP–delivered groups, the unstabilized H5 nanoparticle had lower TX24 HAI ($P < 0.0001$) and neutralizing activity ($P < 0.01$ for all groups except H5-foldon at week 2; $P < 0.0001$ at week 6) than the other groups (Fig. 3, B to F). By contrast, the mRNA-LNP–delivered stabilized nanoparticle elicited significantly higher TX24 HAI titers than all other groups ($P < 0.0001$, Fig. 3D). Postprime, among both protein- and mRNA-LNP–delivered immunogens, the highest TX24 neutralizing activity was seen in the stabilized nanoparticle and both

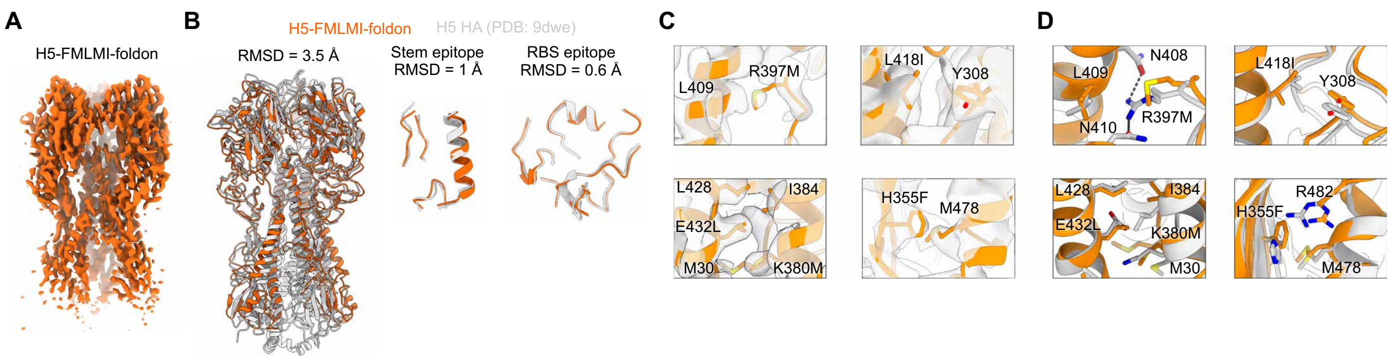

Fig. 2. Cryo-EM of H5-FMLMI-foldon reveals hydrophobic interactions formed by stabilizing mutations. (**A**) Cryo-EM reconstruction of H5-FMLMI-foldon. (**B**) Superimpositions of cryo-EM models for wild-type H5 HA (PDB ID: 9DWE; gray) and H5-FMLMI-foldon (orange) ectodomains (left), central stem epitopes (center), and RBS epitopes (right). (**C**) Close-ups of density for the FMLMI mutations in the H5-FMLMI-foldon reconstruction with the built model. (**D**) Close-ups of the FMLMI mutations and interacting residues in superimposed cryo-EM models of H5-FMLMI-foldon (orange) and wild-type H5 HA (PDB ID: 9DWE; gray).

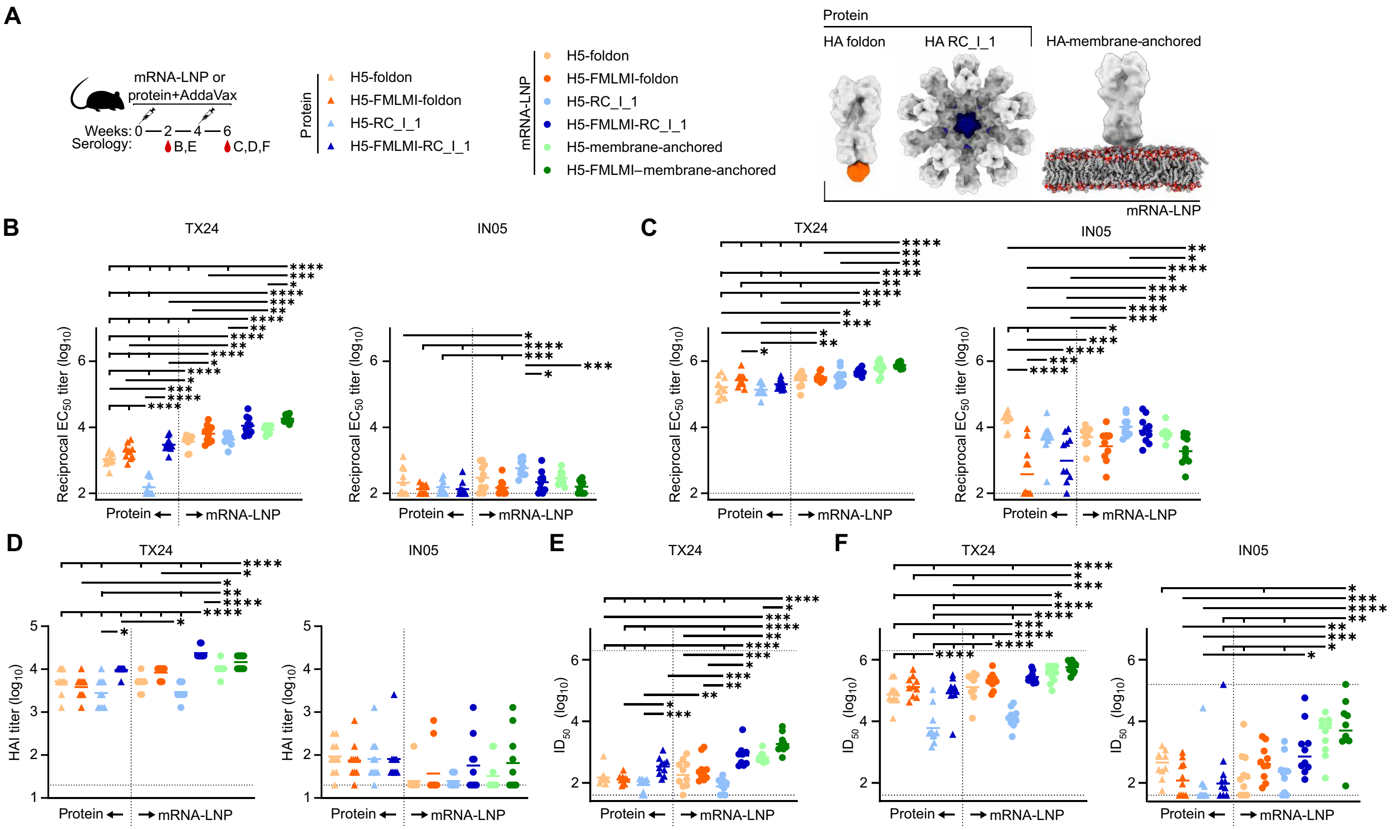

Fig. 3. Antibody responses are elicited in mice immunized with stabilized H5 protein– and mRNA-LNP–delivered constructs. (**A**) H5 mouse immunization schedule and groups. Molecular models of immunogens are not to scale. (**B** and **C**) ELISA binding titers against vaccine-matched TX24 HA-I53_dn5B and vaccine-mismatched IN05 HA-I53_dn5B in immune sera at week 2 (B) and week 6 (C) postvaccination. EC_{50}, half-maximal effective concentration. (**D**) HAI titers in immune sera at week 6. (**E** and **F**) Microneutralization titers in immune sera at week 2 (E) and week 6 (F) postvaccination. Each symbol represents an individual animal, and the geometric mean of each group is indicated by the bar ($n = 9$ or 10 mice per group). Horizontal dashed lines are the lower limits of detection for each assay. Statistical significance was determined using a one-way ANOVA with Tukey's multiple comparisons test; $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, and $****P < 0.0001$. All tick marks are comparing against the rightmost group.

membrane-anchored groups ($P < 0.05$ for all comparisons except to protein-delivered H5-foldon), with the stabilized membrane-anchored group showing significantly higher neutralization than the unstabilized group ($P < 0.05$; Fig. 3, E and F). By contrast with TX24, the unstabilized nanoparticle group had higher IN05 binding titers ($P < 0.05$) but similar IN05 neutralization titers as compared to its stabilized counterpart postprime when delivered as an mRNA-LNP (Fig. 3, B and E). This suggests that the stabilized nanoparticle elicited a greater proportion of functional, vaccine-mismatched antibodies than the unstabilized nanoparticle, perhaps by reducing the proportion of trimer interface–directed antibodies. Together, these findings demonstrate that stabilizing mutations consistently enhanced the elicitation of functional antibody responses in nanoparticle and membrane-anchored constructs.

With stabilizing mutations, the membrane-anchored and nanoparticle groups were observed to have as good or better TX24 HAI ($P < 0.05$ when delivered as mRNA-LNP) and neutralizing activity ($P < 0.05$ at week 2) than the H5-foldon trimer (Fig. 3, D to F). Binding titers against stabilized-stem TX24 were comparable among protein-delivered groups but were higher in the mRNA-LNP–delivered stabilized nanoparticle group than in the membrane-anchored groups ($P < 0.05$). However, these groups showed comparable binding titers against the TX24 HA ectodomain, suggesting that a greater proportion of the antibodies elicited by the stabilized nanoparticle are against the HA stem (Fig. 3C and fig. S3B). However, the greatest proportion of cross-neutralizing antibodies was observed in the stabilized membrane-anchored group, because it had the lowest IN05 binding titers out of all mRNA-LNP groups and the highest IN05 neutralization. Thus, the increased avidity provided by the nanoparticle and membrane-anchored constructs appeared to enhance immune responses as compared with foldon trimers, although the nature of these improvements differed between formats.

mRNA-LNP–delivered constructs overall elicited comparable or improved binding, HAI, and neutralization titers as compared with the same construct delivered as protein. All mRNA-LNP–delivered constructs elicited significantly higher postprime TX24 binding titers than their protein-delivered counterparts ($P < 0.01$; Fig. 3B). In addition, mRNA-LNP–delivered H5-FMLMI-RC_I_1 elicited significantly higher HAI than protein-delivered H5-FMLMI-RC_I_1 ($P < 0.0001$; Fig. 3D). Higher antiscaffold responses were also observed upon vaccine delivery with mRNA-LNP as compared with protein for the nanoparticle groups ($P < 0.0001$), whereas scaffold responses between foldon groups were similar across delivery formats (fig. S3B). Given that the mRNA-LNP delivery of several different

HA constructs resulted in at least comparable and sometimes superior immunogenicity as compared with protein delivery, these results underscore the potential of mRNA-LNP–based vaccine platforms for influenza virus vaccination.

H5 HA–stabilizing mutations increase the proportion of neutralizing RBS-directed responses

We next used nsEM polyclonal epitope mapping (ns-EMPEM) to map the epitopes targeted by vaccine-elicited antibody responses. Purification of vaccine-matched TX24 H5 HA-I53_dn5B bound to polyclonal Fabs purified from serum elicited by the six mRNA-LNP immunogens showed less early-eluting, high-molecular-weight material for all three stabilized H5 HA constructs as compared with their unstabilized counterparts (Fig. 4A).

Immune complex peak fractions collected around 10 to 12 ml were selected for ns-EMPEM analysis. Serum antibodies were observed to bind to previously defined antigenic sites on the HA head (Fig. 4B) (*43*). For the H5-foldon constructs, the vast majority of antibodies elicited by the unstabilized immunogen appeared to target site E, which contains the vestigial esterase domain, whereas the stabilized immunogen induced responses against both site E and the RBS (Fig. 4C). In addition, the H5-foldon immunogen induced a minor population of Fabs bound to the HA monomer, indicative of responses elicited against internal epitopes that disrupt trimerization; however, this was not observed in the H5-FMLMI-foldon group. In the nanoparticle groups, H5-RC_I_1 revealed only Fab-bound HA monomers, with responses against several epitopes along the HA monomer, likely comprising the earlier elution of larger complexes in this unstabilized group. By contrast, all Fabs elicited by H5-FMLMI-RC_I_1 were visualized in complex with trimeric HA, targeting both site E and RBS epitopes. Given that the unstabilized H5 membrane-anchored immune complex SEC profile showed a large shoulder at 10 ml before a slightly larger peak at 11 ml, ns-EMPEM was performed on both fractions. Most complexes in the 10-ml fraction showed multiple Fabs bound to monomeric HA at distinct epitopes, explaining their earlier elution, although a small number of complexes comprising site E– and RBS-directed Fabs bound to trimeric HA were also observed. By contrast, only the latter were observed in the 11-ml fraction. Similarly to the nanoparticle results, stabilization in the H5-FMLMI–membrane-anchored group resulted in no Fab-bound HA monomers and a large proportion of RBS-directed Fabs, in addition to some responses against site E. These data allow us to draw multiple conclusions. First, site E appears to be immunodominant in TX24 HA. Second, stabilization of H5 HA through the introduction of substitutions in HA2 reduces the elicitation of trimer interface–directed antibodies, likely by maintaining the closure of the head domain in vivo as suggested by the lower FluA-20 binding we observed with these constructs in vitro. Last, stabilization of H5 HA promoted elicitation of antibodies targeting the RBS.

To compare the epitopes targeted by neutralizing antibodies in the sera from each group of immunized mice, we used a pseudovirus-based deep mutational scanning (DMS) system. Pseudovirus DMS provides a safe and comprehensive method to measure the effects of

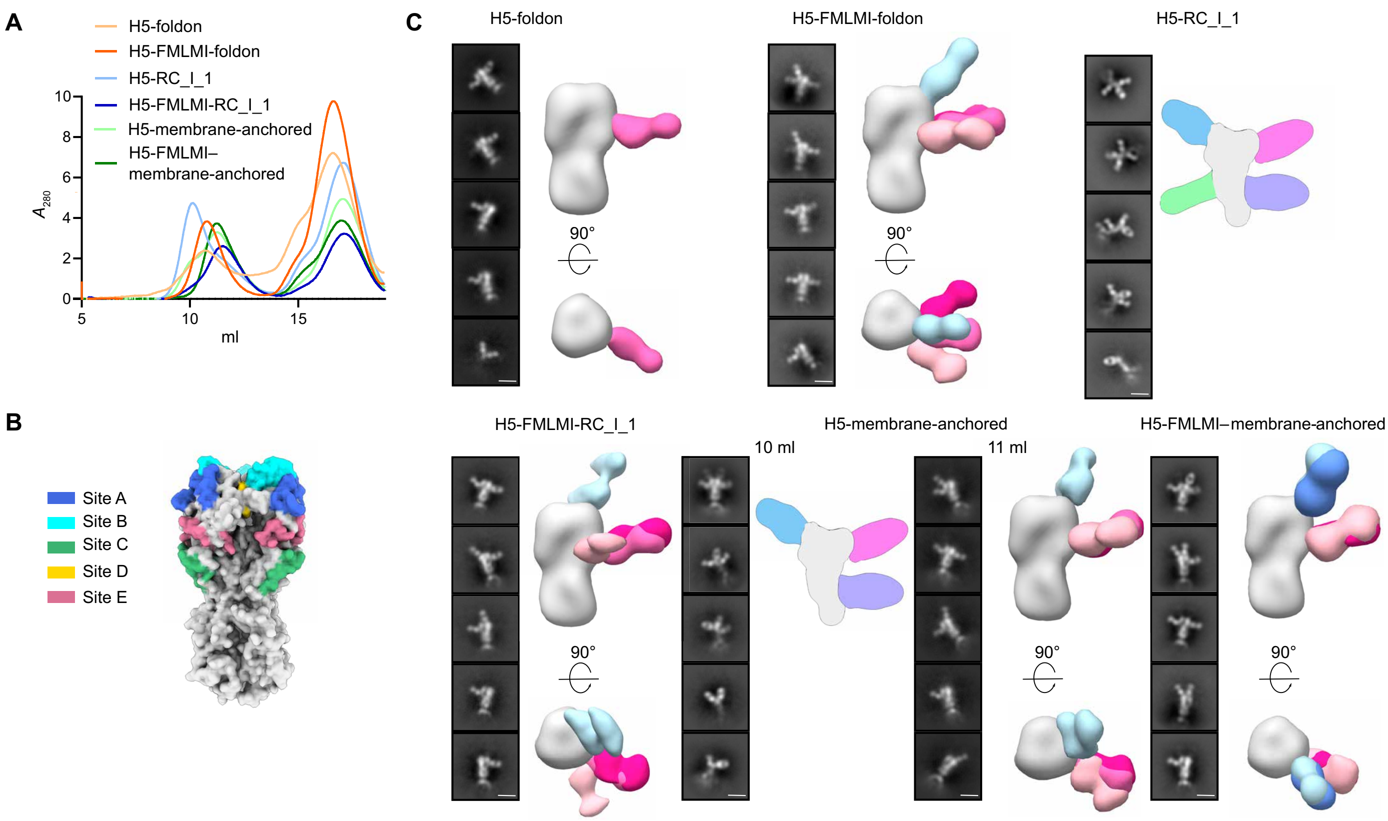

Fig. 4. ns-EMPEM shows that H5-stabilizing mutations promote vaccine-elicited antibody responses against the RBS and noninternal epitopes. (**A**) SEC chromatograms of TX24 HA complexed with polyclonal Fabs from H5 mouse study at week 6. (**B**) Locations of conventional H3 HA antigenic sites as defined in (*43*). (**C**) Representative 2D class averages of purified immune complexes in (A). H5-RC_I_1 and the H5 membrane-anchored 10-ml fraction contain a cartoon schematic of a likely 3D model, whereas all other groups are composite 3D models from ns-EMPEM analysis. Scale bars, 15 nm.

HA mutations on neutralization by using pseudotyped viral particles that can undergo only a single round of cell entry (*11*). A library of pseudoviruses expressing HA variants covering every possible HA ectodomain amino-acid mutation was incubated with sera from multiple animals from each group that received mRNA-LNP–delivered H5 vaccines, and escape mutations that led to decreased serum neutralization were identified (Fig. 5). For all groups except H5-RC_I_1, escape mapping revealed that, regardless of the presence of HA-stabilizing mutations, neutralizing antibodies primarily targeted the A and B epitopes at the top of the HA RBD, including the RBS (Fig. 5, A and B). The sites with the highest escape for these sera were 129, 145, 158, 169, and 189 (H3 numbering). By contrast, when using sera from animals that received H5-RC_I_1, escape mutations were also observed in site E at positions 57, 78, 90, 125, and 276 (Fig. 5, A and B, and table S3). This result is interesting because responses against site E were also visualized in sera from the H5-FMLMI-RC_I_1 group (Fig. 4C). Given that neutralization measured in H5-RC_I_1 sera was low (Fig. 3F), we conclude that this group had low levels of potently neutralizing RBS-directed antibodies, and thus, the presence of more weakly neutralizing site E–directed antibodies was able to be detected in the DMS assay. Although ns-EMPEM data showed that HA stabilization also led to a greater proportion of RBS-targeting antibodies for the membrane-anchored and H5-foldon constructs, no apparent change in neutralizing specificity was present in the DMS

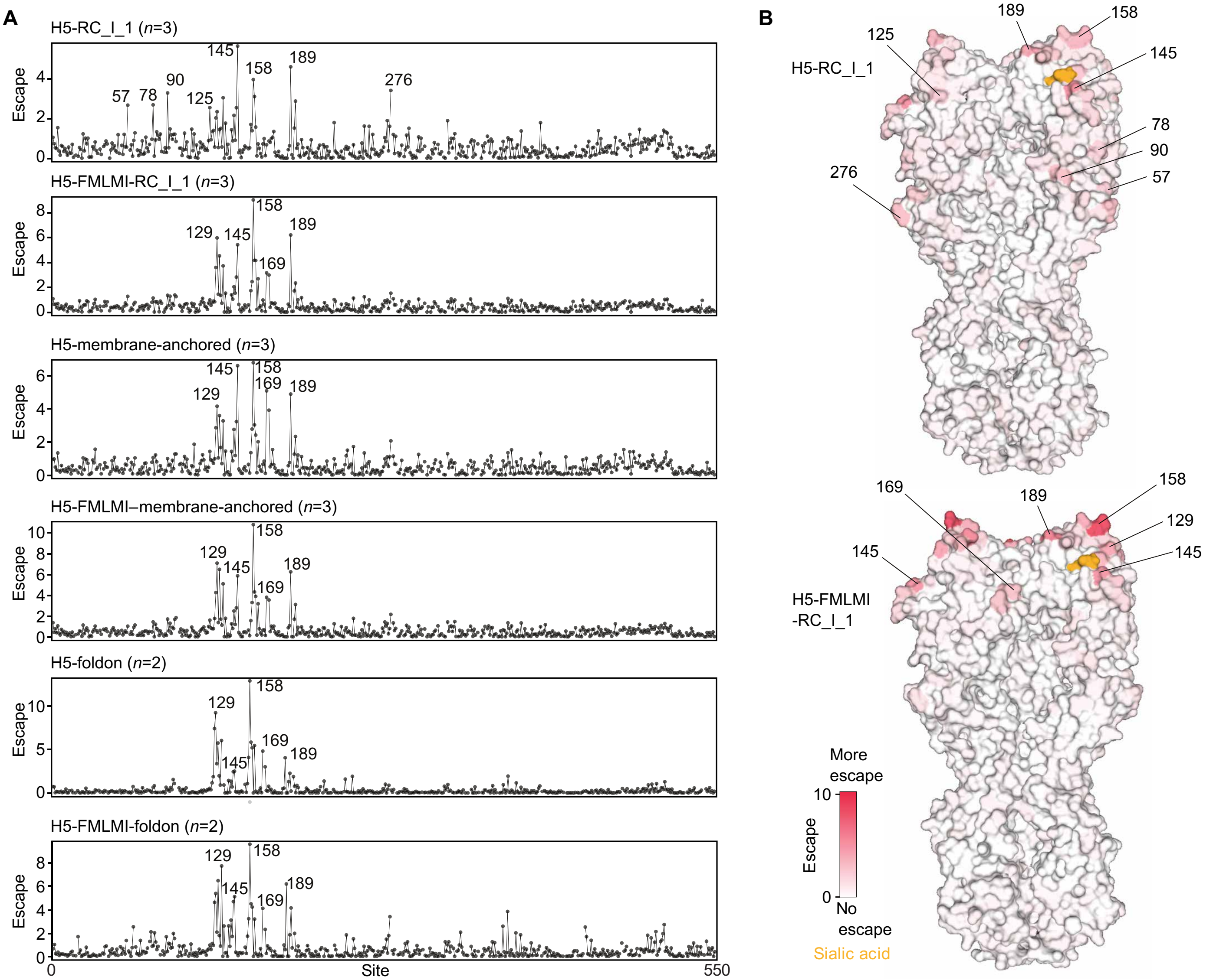

Fig. 5. DMS epitope mapping demonstrates that vaccine-elicited antibodies against the RBS have the highest neutralization potency. (**A**) Escape from serum antibody neutralization for all tolerated amino-acid mutations at each site in HA. Average escape for all animals in each group is shown, with the number of individual animal sera from each group analyzed indicated in parentheses. Escape measurements are proportional to $\log_2$ fold change in neutralization (IC_{50}). See https://dms-vep.org/Flu_H5_American-Wigeon_South-Carolina_2021-H5N1_DMS/htmls/Stabilized_HA_sera_escape_faceted.html (*67*) for individual animal escape measurements and interactive plots. The $n = 2$ or 3 above each plot represents the number of animals used to calculate escape. (**B**) Average serum escape per position for animals from the H5-RC_I_1 and H5-FMLMI-RC_I_1 groups overlaid on an HA surface representation (PBD ID: 4KWM). Numbering is based on mature H3 HA.

data for these stabilized and unstabilized HAs. For all constructs except H5-RC_I_1, it is likely that sufficient antibodies targeting residues adjacent to the RBS were elicited, resulting in most neutralizing activity being directed toward this region, irrespective of HA stabilization.

H5 HA mRNA-LNP constructs protect against homologous challenge in mice

To gauge the ability of H5 RC_I_1 and membrane-anchored constructs to confer protection, we challenged groups of 10 mice with five times the half-maximal tissue culture infection dose ($TCID_{50}$) of vaccine-matched H5N1 A/Texas/37/2024 virus (Fig. 6A). Both H5 membrane-anchored groups and H5-FMLMI-RC_I_1 resulted in complete protection, with no loss in body weight up to 14 days after challenge (Fig. 6, B and C). By contrast, the H5-RC_I_1 group resulted in one mouse succumbing to disease and a mean weight loss of up to 7.2%, whereas all mice receiving the empty LNP control succumbed to the infection. These results are in agreement with the HAI and neutralization data (Fig. 3, D and F), demonstrating similarly high protection in the stabilized and unstabilized membrane-anchored and stabilized nanoparticle groups.

DISCUSSION

Here, we showed that stabilizing mutations can enhance thermal stability and improve the quality of vaccine-elicited antibody responses in mice across multiple H5 HA immunogens delivered both as protein and LNP-formulated mRNA. Although all soluble HA constructs formed trimeric, prefusion HA0 molecules by nsEM, the stabilizing FMLMI mutations diminished nonneutralizing FluA-20 binding in the foldon trimer, RC_I_1 nanoparticle, and membrane-anchored groups, suggesting more efficient closed trimer formation in the head domains of these constructs. This resulted in a reduction in internally directed antibodies elicited by the stabilized immunogens in vivo, with a simultaneous boosting of neutralizing RBS-directed responses. We have observed this phenomenon previously, where stabilized, closed trimeric RBD nanoparticles elicited higher neutralizing responses across several H1 strains than open monomeric RBD nanoparticles, and only the latter induced high amounts of internally directed antibodies (*44*). In addition to epitope (in)accessibility, the improved stability we observed in the FMLMI immunogens may also be a key driver of their enhanced immunogenicity. Previous studies using human immunodeficiency virus (HIV) envelope (*45*, *46*) and respiratory syncytial virus (RSV) fusion (*25*) immunogens showed positive correlations between increasing physical stability and vaccine-elicited neutralizing responses. Moreover, although we did not explore antigen retention studies here, recent work has shown that proteolytic degradation in lymph nodes is a major hurdle for subunit vaccines to overcome to reach follicles where adaptive immune responses are generated (*47*). Thus, the correlation between physical stability and enhanced immunogenicity may result from the extension of antigen integrity in vivo. Although both ns-EMPEM and DMS showed binding to multiple epitopes, we observed by ns-EMPEM a high amount of site E–binding mAbs across three different H5 HA immunogen formats delivered via mRNA-LNP. The immunodominance of this site in this naïve BALB/c mouse model mirrors the targets of anti-HA B cell hybridomas generated early in the response to H1N1 infection in the same type of mice (*48*). In that study, responses to antigenic sites A and B near the RBS only emerged later during the course of infection. By contrast with infection, where newly produced virions produce a constant supply of new antigen, vaccination in the present study was limited to the delivery of new antigen at two discrete time points. Although speculative, this further supports the idea that the stabilizing FMLMI mutations may prolong intact antigen availability in vivo to not only generate initial responses against the immunodominant site E but also potentially drive an increase in later responses against the more neutralizing yet subdominant RBS (*49*).

The stabilizing FMLMI mutations enhanced immunogenicity regardless of construct design and delivery modality, highlighting the utility of incorporating these mutations into any type of HA-based influenza vaccine. This result is particularly timely given the pandemic threat of H5 HPAI viruses currently circulating widely in birds and mammals. Incorporating stabilizing mutations into H5 HA vaccine platforms that can be rapidly produced and scaled up could be an important pandemic preparedness measure (*50*). Although stabilizing mutations were only tested in the context of H5 HA in this study, Milder *et al.* (*28*) showed substantial stability improvements by incorporating similar mutations across HAs from type A strains. This further suggests that vaccines targeting any influenza A subtype may be improved by incorporating these universal stabilizing mutations.

More broadly, antigen multimerization in the mRNA-LNP–delivered stabilized nanoparticle and both membrane-anchored

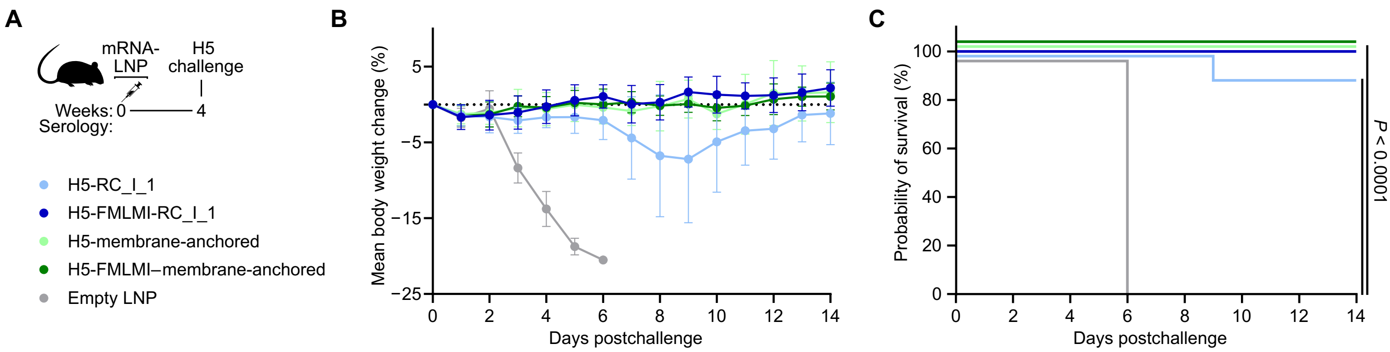

Fig. 6. Vaccination of mice with H5 mRNA-LNP–delivered constructs confers protection against vaccine-matched H5N1 A/Texas/37/2024 challenge. (**A**) H5 mouse immunization and H5N1 A/Texas/37/2024 challenge schedule and groups (n = 10 mice per group). (**B** and **C**) Weight loss (B) and probability of survival (C) after H5N1 A/Texas/37/2025 challenge. Each point and error bar in (B) represent the means ± SEM for groups of mice. The multiple Kaplan-Meier curves in (C) were compared using Mantel-Cox log-rank test with Bonferroni correction.

formats enhanced immunogenicity compared with soluble HA trimers, a finding that has been observed previously (*17*). This effect is further underscored by the observation that H5-FMLMI-foldon had considerably higher expression than H5-FMLMI-RC_I_1 when produced in HEK293F cells, suggesting that the improved immunogenicity of the H5-FMLMI-RC_I_1 nanoparticle is not simply a result of increased protein expression for mRNA-LNP–delivered immunogens. Furthermore, H5-FMLMI-RC_I_1 elicited higher HAI titers than all other groups when delivered via mRNA-LNP, whereas HAI titers were similar across all protein-delivered groups. Together, these results highlight the potentially synergistic benefit of combining multivalent antigen display with mRNA-LNP–delivered vaccines. However, the prominent instability of H5-RC_I_1, which was less stable than the H5-foldon, suggests that the RC_I_1 nanoparticle scaffold is suboptimal and may constrain the immunogenicity of RC_I_1-based nanoparticle immunogens. Despite this scaffold instability, the stabilized H5-FMLMI-RC_I_1 immunogen elicited neutralizing responses comparable to those of the H5-FMLMI–membrane-anchored immunogen. It follows from these data that the display of stabilized HA on a better nanoparticle scaffold may further improve immunogenicity. Together, our data underscore that, for protein nanoparticle or mRNA-launched nanoparticle vaccines, several features of both the antigen and scaffold, including conformation, stability, expression, and secretion, should be maximized to optimize vaccine efficacy.

One limitation of this study is that the immune responses observed in mice will likely differ from those elicited in humans. The mouse B cell repertoire has shorter heavy-chain complementary determining region 3 loops on average, with substantially different amino acid usage, as compared with the human repertoire (*51*). Moreover, humans have complex immune histories of repeated influenza vaccinations and infections that will influence the resulting immunogenicity of any new influenza vaccine (*52*). Additional limitations are inherent to each of the epitope mapping approaches. For example, although EMPEM is a measure of antibody binding, it is biased toward antibodies with greater affinities and abundance, and it cannot quantitatively assess either parameter. As a complementary epitope-mapping technique, the DMS conducted here measures antibody neutralization. In H5-foldon sera, DMS detected only strongly neutralizing site A– and site B–directed antibodies, whereas EMPEM revealed only site E–directed antibodies, likely reflecting their greater abundance or affinity but comparatively weaker neutralization.

In conclusion, this work underscores the potential to improve immunogenicity by incorporating stabilizing mutations across an array of H5 HA vaccine platforms. The efficacy of these vaccines when delivered as mRNA-LNPs, coupled with their fast manufacturing timelines, makes them particularly amenable against H5N1 viruses with a high pandemic potential.

MATERIALS AND METHODS

Study design

The main objective of this study was to evaluate the impact of introducing stabilizing mutations in H5 HA on immunogenicity. Multiple HA constructs were tested both with and without stabilizing mutations in a preclinical mouse model. Given that multimerization is a known driver of immunogenicity, we compared these stabilizing mutations in the context of both soluble trimer and protein nanoparticle immunogens. In addition, we assessed these immunogens both as protein- and mRNA-LNP–delivered vaccines. By immunizing as protein, we were able to give the same amount of HA across all groups, allowing us to evaluate vaccine constructs independent of expression level. By contrast, delivery with a dose-matched amount of mRNA-LNP allowed us to evaluate these constructs in the rapidly manufacturable and highly scalable mRNA-LNP platform. Given that membrane-anchored HA is being widely evaluated as an mRNA-LNP influenza vaccine, we included both stabilized and unstabilized versions of membrane-anchored H5 HA. Two immunizations were administered (weeks 0 and 4) to analyze immune responses after prime (week 2) and after boost (week 6). ELISA binding, HAI, and microneutralization using both vaccine-matched and vaccine-mismatched H5 strains were performed to assess functionality, potency, and breadth of the vaccine-elicited immune responses. ns-EMPEM was used to visualize antibody binding epitopes, and DMS was used to evaluate epitopes targeted by neutralizing antibodies.

All groups consisted of 10 mice to maximize statistical power to assess differences between groups. However, blood from one mouse in the mRNA-LNP–delivered H5-RC_I_1 group was unable to be drawn at week 6, so there are only data from nine mice in this group at this time point. Animals were randomly assigned to groups, and no blinding was implemented. Numbers of biological and technical replicates are provided in either the figure legends or Materials and Methods. Although all HA numbering in this paper is based on H3 HA numbering, conversions to other HA numbering schemes can be found here: https://dms-vep.org/Flu_H5_American-Wigeon_South-Carolina_2021-H5N1_DMS/numbering.html (*53*).

Animal experiments were conducted in accordance with the University of Washington's Institutional Animal Care and Use Committee (IACUC protocol 4470-01). The University of Washington's animal use program is accredited by the Association for Assessment and Accreditation of Laboratory Animal Care (reference assurance no. 000523).

Gene expression and protein purification

All HA constructs used in this study were codon optimized for human cell expression and made in the CMV/R vector (*54*) by Genscript with a C-terminal hexahistidine affinity tag. PEI MAX was used for transient transfection of HEK293F cells. After 4 days, mammalian cell supernatants were clarified via centrifugation and filtration. HA protein immunogens and HA foldons were all purified using immobilized metal affinity chromatography. One milliliter of Ni^{2+}-sepharose Excel was added per 100 ml of clarified supernatant along with 5 ml of 1 M tris, pH 8.0, and 7 ml of 5 M NaCl and left to batch bind while shaking at room temperature for 30 min. Resin was then collected in a gravity column and washed with five column volumes of 50 mM tris, pH 8.0, containing 500 mM NaCl and 20 mM imidazole. Protein was eluted using 50 mM tris, pH 8.0, containing 500 mM NaCl and 300 mM imidazole. Further purification was done using SEC on a Superose 6 Increase 10/300 gel filtration column equilibrated in 25 mM tris, pH 8.0, containing 150 mM NaCl and 5% glycerol. HA-ferritin nanoparticles used in HAI assays were purified using lectin affinity chromatography, followed by SEC on a Superose 6 Increase 10/300 gel filtration column (*55*).

After purification, the nanoparticle quality and concentration were measured by ultraviolet-visible spectroscopy. Nanoparticle polydispersity and purity post–freeze-thaw were then assessed using SDS–polyacrylamide gel electrophoresis, DLS, and nsEM. Last, endotoxin levels were measured using the *Limulus* amebocyte lysate assay, with all immunogens used in animal studies containing less than 100 EU/mg in the final dose. Final immunogens were flash-frozen using liquid nitrogen and stored at −80°C.

NanoDSF

All proteins were formulated at 0.5 mg/ml in 25 mM tris, pH 8.0, containing 150 mM NaCl and 5% glycerol and then mixed at nine volumes to one volume of 200× concentrate SYPRO orange (Thermo Fisher Scientific) diluted in the same buffer. NanoDSF to determine melting temperatures was carried out on an UNcle (UNchained Labs) by measuring the integration of fluorescence emission spectrum during a thermal ramp from 25° to 95°C, with a 1°C increase in temperature per minute.

SDS-page

Twelve microliters of protein ranging from 0.5 to 3 mg/ml protein was loaded onto Criterion TGX stain-free, any-kilodalton, 15-well gels (Bio-Rad), in addition to 5 μl of Plus unstained protein standards (Bio-Rad). Gels were run with 250 V applied for 19 min and subsequently imaged with a gel imager using the stain-free protocol.

Negative-stain electron microscopy

Three and a half microliters of 70 μg/ml nanoparticles was applied to glow-discharged 400-mesh carbon-coated grids (Electron Microscopy Sciences) and stained with 2% (w/v) uranyl formate. Data were collected using EPU 2.0 on a 120-kV Talos L120C transmission electron microscope (Thermo Fisher Scientific) with a BM-Ceta camera. CryoSPARC (*56*) was used for CTF correction, particle picking and extraction, and 2D classification.

Dynamic light scattering

DLS was carried out on an UNcle (UNchained Labs) at 25°C. Ten acquisitions of 5 s each were acquired for each spectrum. The protein concentration (ranging from 0.1 to 1 mg/ml) and buffer conditions were accounted for in the software.

Mass photometry

MP was carried out in a TwoMP (Refeyn) Mass photometer. A 12-well gasket was placed on each slide. Ten microliters of buffer was added to one well of this gasket, and the camera was brought into focus after orienting the laser to the center of the sample well. Next, 2 μl of H5-FMLMI-RC_I_1 at 120 nM nanoparticle was added to this droplet, and 1-min videos were collected with a large field of view in AcquireMP. Ratiometric contrast values for individual particles were measured and processed into mass distributions with DiscoverMP. A sample of 20 nM β-amylase, consisting of monomers (56 kDa), dimers (112 kDa), and tetramers (224 kDa) in equilibrium, was used to arrive at a mass calibration, thereby allowing contrast values to be converted into mass values. Distributions and Gaussian fit were plotted using the seaborn Python library (*57*).

Biolayer interferometry

BLI was carried out using an Octet Red 96 system at 25°C with 1000-rpm shaking. AR2G biosensors were activated in 10 mM sulfo-*N*-hydroxysuccinimide and 10 mM carbodiimide 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide. Anti-HA antibodies were diluted in acetate buffer to a final concentration of 20 μg/ml before loading onto activated AR2G biosensors (Sartorius). For assaying CR9114, 12D03 was used as the loading antibody at pH 3.5, whereas CR9114 was used as the loading antibody at pH 5.0 for assaying all other antibodies. After quenching in 1 M ethanolamine, HA proteins diluted in 500 nM kinetics buffer [phosphate-buffered saline (PBS) with 0.5% serum bovine albumin and 0.01% Tween] were then loaded, followed by association of the assayed antibody at 20 μg/ml and subsequent dissociation in kinetics buffer. All loading, association, and dissociation steps were carried out for 300 s, with 120-s baseline steps in kinetics buffer in between.

Flow cytometry

HEK293F cells were transiently transfected with HA membrane-anchored constructs as described above. After a wash with PBS, Zombie Aqua live/dead viability dye at 1:1000 (BioLegend, no. 423101) and 1 μM biotinylated FluA-20 were added to cells and left to incubate for 15 min. Cells were again washed and subsequently stained with streptavidin, R-phycoerythrin conjugate (SAPE; Thermo Fisher Scientific, no. S866), and 200 nM Alexa Fluor 647–conjugated (Thermo Fisher Scientific, no. A88068) CR9114 for another 15 min. After a final wash, flow cytometry was performed using an Attune flow cytometer (Thermo Fisher Scientific). The live/singlet populations were used in final binding plots.

Cryo-EM sample preparation, data collection, and processing

Cryo-EM samples were prepared on glow-discharged QUANTIFOIL 1.2/1.3 holey-carbon grids (Electron Microscopy Sciences). A protein sample at 1 mg/ml was applied to the grid, incubated for 30 s, and then blotted using Whatman filter paper; this process was performed twice, followed by a third application of sample and final blotting and plunge freezing in liquid-ethane using a Vitrobot (FEI). Data were acquired on a Titan Krios operating at 300 kV equipped with a K3 Summit Direct Detector and a Quantum GIF energy filter (Gatan) operating in zero-loss mode with a 20-eV slit width. Movies were collected at both 0° and 30° in counting mode, collecting 100 frames with a total dose of 60 $e^{-}/Å^{-2}$, with a defocus range of −0.5 to −2 μm. Automated data collection was carried out using SerialEM (*58*) at a nominal magnification of 105,000× with a pixel size of 0.843 Å. All data processing was carried out using cryoSPARC (*56*). Movie frame alignment and dose-weighting summation were performed using Patch Motion, and CTF estimation was done using patch CTF. Particles were first picked using Blob Picker, and these were used to generate initial 2D averages. Two more rounds of 2D classification were performed to select all well-aligned, single HA views, followed by a final selection of only HA side views. Top views of HA were excluded because of their preferred orientation and, thus, potential damage at the air-water interface. An initial model was generated by ab initio reconstruction with C1 symmetry, followed by an initial homogeneous 3D refinement with C3 symmetry. A solvent mask around the HA trimer was then used to exclude noise in a 3D classification without alignment and in all subsequent steps. The final homogeneous refinement of the 3D class with the most particles resulted in the best-resolved map at 3.2 Å and was used to build an atomic model.

Model building and refinement

An AlphaFold3 (*59*) model was docked into the cryo-EM map as a rigid body using Chimera (*60*). A single *N*-acetylglucosamine at each nonclashing glycan sequon was added using GlycoShape (*61*). An initial refinement of one HA protomer into the cryo-EM density was performed with ISOLDE (*62*). C3 symmetric copies of this fit protomer were then generated, and this HA trimer model was relaxed into the cryo-EM density again in ISOLDE. A final real-space refinement with grid searching, Ramachandran, and rotamer restraints all turned off and "use starting model as reference" turned on was then performed in Phenix (*63*). Flexible loops comprising residues 324 to 337, as well as the foldon domain, were not resolved in the cryo-EM density, so these loops were not modeled.

mRNA design and production

The H5 HA–encoding mRNAs were produced as described (*21*). Briefly, the H5 HA sequences were codon optimized, synthesized, and cloned into an mRNA production plasmid. The mRNA was transcribed to contain a 101 nucleotide–long poly(A) tail. N1-methylpseudouridine-5′-triphosphate (m1Ψ-5′-triphosphate) (TriLink no. N-1081) instead of uridine-5′-triphosphate was used to generate modified nucleoside-containing mRNA. Capping of the in vitro–transcribed mRNAs was performed cotranscriptionally using the trinucleotide cap1 analog CleanCap (TriLink no. N-7413). mRNA was purified by cellulose (Sigma-Aldrich, no. 11363-250G) purification. The mRNAs were analyzed by agarose gel electrophoresis and were stored frozen at −20°C.

LNP encapsulation of mRNA

mRNA-LNPs with a composition similar to that described previously (*21*) were produced, containing cationic ionizable lipid and PEGylated lipid proprietary to Acuitas Therapeutics. The lipids and formulation are described in patent application WO 2017/004143A1 (*64*). Briefly, ionizable lipids, distearoylphosphatidylcholine, cholesterol, and PEGylated lipids were dissolved in ethanol and rapidly mixed with mRNA diluted in an aqueous citrate pH 4 buffer. Formulations were then dialyzed overnight with PBS, concentrated to target 1 mg/ml with a cryoprotectant, filtered, vialed, and then frozen at −80°C. mRNA-LNPs were characterized at Acuitas Therapeutics for their size, polydispersity using a Malvern Zetasizer (Malvern Panalytical), encapsulation efficiency using the standard Ribogreen assay (Thermo Fisher Scientific), and mRNA content by ultraperformance liquid chromatography analysis (Waters).

Immunizations for serology

Female BALB/c mice were purchased from Envigo (order code 047) at 7 weeks of age and were housed in a specific pathogen–free facility within the Department of Comparative Medicine at the University of Washington, Seattle. Animals had access to standard rodent chow and water ad libitum and were maintained in an ABSL1 housing room with environmental parameters set to 68° to 79°F, 30 to 70% humidity, and a dark/light cycle of 10 hours of dark and 14 hours of light. At 8 weeks of age, mice were immunized subcutaneously in the inguinal region with 21.8 pmol of protein in a 50% (v/v) mixture of AddaVax adjuvant (InvivoGen) or intramuscularly with 1 μg of mRNA-LNP, formulated in 100 μl of PBS (50 μl in each quadricep) at weeks 0 and 4. For mice receiving intramuscular injections, preemptive buprenorphine HCl (0.05 mg/kg; NDC 42023-179-05) was administered subcutaneously to prevent pain. For sera, animals were bled using the submental route at week 0 and 2 weeks after immunization. Whole blood was collected in serum separator tubes (BD no. 365967) and rested for 30 min at room temperature for coagulation. Tubes were then centrifuged for 10 min at 2000*g*. Serum was collected and stored at −80°C until use.

ELISA

H5-I53_dn5B trimers were added to 96-well half-area high-binding plates (Corning) at 2.0 μg/ml with 25 μl per well and incubated overnight at 4°C. Blocking buffer composed of tris-buffered saline with Tween [25 mM tris, pH 8.0, 150 mM NaCl, and 0.05% (v/v) Tween 20] with 5% nonfat milk was then added at 100 μl per well and incubated for 1 hour. Next plates were washed, with all washing steps consisting of washing three times with tris-buffered saline with Tween using a robotic plate washer (BioTek). Fivefold serial dilutions of serum starting at 1:100 were made in blocking buffer, added to plates at 25 μl per well, and incubated for 1 hour. Plates were washed again before addition of 25 μl per well of anti-mouse horseradish peroxidase–conjugated secondary antibody (Cell Signaling Technology) diluted 1:2000 in blocking buffer and incubated for 30 min. All incubations were carried out with shaking at room temperature. Plates were washed a final time, and then 50 μl per well of TMB (3,3′,5′,5-tetramethylbenzidine; SeraCare) was added for 5 min, followed by quenching with 50 μl per well of 1 M HCl. Reading at 450-nm absorbance was done on an Epoch plate reader (BioTek).

HAI

Serum was inactivated using receptor-destroying enzyme II (RDE II; Seiken) in PBS at a 3:1 ratio of RDE to serum for 16 hours at 37°C, followed by 40 min at 56°C. Inactivated serum was serially diluted twofold in PBS in V-bottom plates at 25 μl per well. Twenty-five microliters of HA-ferritin nanoparticles at four hemagglutinating units was then added to all wells and incubated at room temperature for 30 min (*35*). Last, 50 μl of 10-fold diluted turkey or horse red blood cells (Lampire) in PBS was added to each well. Hemagglutination was left to proceed for at least 1 hour before recording the HAI titer.

Reporter-based microneutralization assay

Replication-restricted, rewired reporter (R4ΔPB1) influenza H5N1 viruses were generated as previously described (*37*, *65*). Briefly, each R4 virus was rescued using a modified PB1 segment, which was engineered by replacing PB1 with the coding region of H5 HA flanking the PB1 genome packaging signal on each end. The H5 HA sequences were modified by introducing synonymous mutations to inactivate the HA genomic packaging signals and replacing the polybasic cleavage site with a monobasic motif, allowing proteolytic HA processing to occur only in the presence of exogenous trypsin. The HA segments were prepared by replacing the HA coding region with the fluorescent reporter TdKatushka2 fused with a nuclear localization signal at the C terminus flanking the HA genomic packaging signals of A/Puerto Rico/8/1934 (GenBank MW298214.1) on each end. H5N1 R4ΔPB1 viruses (A/Texas/37/2024 and A/Indonesia/05/2005) were rescued by reverse genetics and propagated in MDCK-SIAT-PB1 cells in the presence of tosyl phenylalanyl chloromethyl ketone–treated trypsin (1 μg/ml; Sigma-Aldrich) at 37°C. Virus stocks were stored at −80°C and titrated before use in the assay. All procedures including reverse genetics techniques used to generate the replication-restricted reporter influenza viruses were reviewed and approved by the National Institutes of Health (NIH) Institutional Biosafety Committee.

Mouse immune sera were treated with RDE (RDE II; Denka Seiken) at a 1:3 ratio (sera:RDE_II), heat inactivated, and then serially diluted with Opti-MEM before addition of pretitrated H5N1 R4ΔPB1 virus. After incubating at 37°C for an hour, prewashed MDCK-SIAT-PB1 cells (1×10^6 cells/ml) were added to the serum-virus mixtures and transferred to 384-well plates in quadruplicate

(25 μl per well). Fluorescence units were counted after overnight incubation at 37°C using a Celigo image cytometer (Nexcelom Biosciences). The target virus control range for this assay is 500 to 2000 fluorescence units per well. The percent neutralization was calculated for each well by constraining the virus control (virus and cells) and the cell-only control (no virus) as 0 and100% neutralization, respectively. A seven-point neutralization curve was plotted against serum dilution for each sample, and a four-parameter nonlinear fit was generated using GraphPad Prism (version 10.1.0, GraphPad) to calculate the 50% inhibitory dilution (ID_{50}) of serum.

ns-EMPEM
Fifty microliters of serum from each of the 10 mice per group from the immunization-only study was pooled and then mixed with an equal volume of acetate buffer, pH 5.0, and incubated overnight with 0.5 ml of packed protein G agarose resin (Thermo Fisher Scientific). The flowthrough was removed using gravity purification, and resin was washed with 20 column volumes of PBS. Immunoglobulin G (IgG) antibodies were eluted by incubating resin for 20 min with one column volume of 0.1 M glycine, pH 2.5, repeated twice. Elutions were neutralized with 1 M tris, pH 8.0, to a final concentration of 50 mM. IgG antibodies were then buffer exchanged into PBS and concentrated to 200 μl for digestion into Fabs. Fab digestion was carried out by adding 200 μl of freshly made 2× digestion buffer (40 mM $NaPO_4$, pH 6.5, 20 mM EDTA, and 40 mM cysteine) and 300 μl of papain resin in 300 μl of 1× digestion buffer, and the sample was incubated with shaking at 37°C for 16 hours. The papain digestion reaction was centrifuged, and the supernatant containing Fabs was collected and filtered. Papain resin was then washed with one column volume of 20 mM tris, pH 8.0, and this supernatant was added to the first papain digestion supernatant. Fabs were then concentrated to 50 μl and mixed at a 50-fold molar excess to the H5-I53_dn5B trimer. Samples were then incubated for 16 to 20 hours at room temperature with gentle rocking. Immune complexes were purified by SEC on a Superdex 200 Increase 10/300 GL column and used in electron microscopy.

nsEM data were collected using EPU 2.0 on a 120-kV Talos L120C transmission electron microscope (Thermo Fisher Scientific) with a BM-Ceta camera. Data processing was done in cryoSPARC (*56*), starting with CTF correction, particle picking, and extraction. Three rounds of 2D classification were done, keeping only classes that had either the HA monomer or trimer with bound Fabs. 3D models for these immune complexes were then generated using ab initio 3D reconstruction and heterogeneous refinement without imposing any symmetry. Classes that had clear, trimeric HA density and were representative of the diversity of Fab binding in each sample of polyclonal serum without redundancy were then separately subjected to a 3D refinement.

Deep mutational scanning
H5 HA deep mutational scanning libraries were produced on the A/American Wigeon/South Carolina/USDA-000345-001/2021 strain background, as described previously (*11*). The safety of deep mutational scanning experiments is ensured by using pseudoviruses that can only undergo a single round of infection, do not encode any other viral genes than HA, and require lentiviral helper plasmids for virus production. The A/American Wigeon/South Carolina/USDA-000345-001/2021 strain differs from the A/Texas/37/2024 strain by two mutations (L122Q and T199I; H3 numbering) in the HA gene. To measure how HA mutations affect serum neutralization, deep mutational scanning libraries were incubated for 45 min at 37°C with serum dilutions from the immunization-only study that neutralized between 80 and 99% of the library viruses. After incubation with sera, library mixtures were used to infect 293T cells. Fifteen hours postinfection, pseudoviral genomes were recovered and prepared for Illumina sequencing. One biological replicate was performed for each serum sample. Before use, all sera were treated with RDE for 2 hours at 37°C and subsequently heat inactivated at 56°C for 30 min. A biophysical model implemented in dms-vep-pipeline-3 was used to calculate escape caused by each mutation in the libraries (*66*). Interactive plots for serum escape data are available at https://dms-vep.org/Flu_H5_American-Wigeon_South-Carolina_2021-H5N1_DMS/htmls/Stabilized_HA_sera_escape_faceted.html (*67*). The analysis pipeline used to model serum escape is deposited to Zenodo under DOI 10.5281/zenodo.16740862 (*68*) and is available on GitHub: https://github.com/dms-vep/Flu_H5_American-Wigeon_South-Carolina_2021-H5N1_DMS (*69*). The plots in Fig. 5 show the total escape caused by all tolerated amino acids at each site, averaging over mice in each group.

Immunization and virus challenge
Female BALB/cAnNHsd mice between 4 and 6 weeks of age were purchased from Envigo and were housed in either a conventional or specific pathogen–free animal facility on a 12-hour light/dark cycle under ambient conditions with free access to food and water. Animals were randomly assigned to experimental groups. All experiments were done in accordance with approved guidelines, regulations, and protocols, as determined and approved by the Institutional Animal Care and Use Committee at Bioqual Inc. The research facility is AAALAC (American Association for Accreditation of Laboratory Animal Care) International accredited, and standards for all animal care (acquisition, breeding, and experimental protocols), biosafety, and personnel occupational health and safety conform to all federal, state, and local regulations. Mice were immunized once on day 0 of the study (50 μl per hindlimb, 1 μg of total mRNA per mouse) and then challenged with 10 times the half-maximal lethal dose (LD_{50}) (5 $TCID_{50}$ per dose) of H5 A/Texas/37/2024 (CDC Seed Stock, lot no. 3016006625) on day 28. Virus challenge was initiated in a high containment laboratory (ABSL-3) by intranasal inoculation, applying 50 μl by a micropipette to the nares while mice were anesthetized with isoflurane. Mice were weighed once daily and checked twice daily for 14 days. Clinical signs of disease (changes in appearance and behavior and neurological signs) were noted at each check. Mice that were either moribund or suffering respiratory distress, a ≥20% loss in body weight from the baseline, or severe neurological signs (seizures or hindlimb paralysis) were euthanized immediately using carbon dioxide. Euthanasia was performed using procedures consistent with American Veterinary Medical Association guidelines.

Statistical analysis
Individual-level data are presented in data file S1. Multigroup comparisons were performed using the Brown-Forsythe one-way analysis of variance (ANOVA) test followed by Tukey's multiple comparisons test in Prism 9 (GraphPad). DMS significance was calculated using the Mann-Whitney *U* test. Differences were considered significant when *P* values were less than 0.05. For the probability of survival data, multiple Kaplan-Meier curves were compared using the Mantel-Cox log-rank test with Bonferroni correction. For all statistical analyses, $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, and $^{****}P < 0.0001$.

Supplementary Materials

The PDF file includes:
Figs. S1 to S3
Tables S1 to S3
Legend for data file S1

Other Supplementary Material for this manuscript includes the following:
Data file S1
MDAR Reproducibility Checklist

REFERENCES AND NOTES

1. D. J. Alexander, An overview of the epidemiology of avian influenza. *Vaccine* **25**, 5637–5644 (2007).
2. V. Caliendo, N. S. Lewis, A. Pohlmann, S. R. Baillie, A. C. Banyard, M. Beer, I. H. Brown, R. A. M. Fouchier, R. D. E. Hansen, T. K. Lameris, A. S. Lang, S. Laurendeau, O. Lung, G. Robertson, H. van der Jeugd, T. N. Alkie, K. Thorup, M. L. van Toor, J. Waldenström, C. Yason, T. Kuiken, Y. Berhane, Transatlantic spread of highly pathogenic avian influenza H5N1 by wild birds from Europe to North America in 2021. *Sci. Rep.* **12**, 11729 (2022).
3. R. J. Webby, T. M. Uyeki, An update on highly pathogenic avian influenza A(H5N1) virus, clade 2.3.4.4b. *J Infect Dis* **230**, 533–542 (2024).
4. E. R. Burrough, D. R. Magstadt, B. Petersen, S. J. Timmermans, P. C. Gauger, J. Zhang, C. Siepker, M. Mainenti, G. Li, A. C. Thompson, P. J. Gorden, P. J. Plummer, R. Main, Highly pathogenic avian influenza A(H5N1) clade 2.3.4.4b virus infection in domestic dairy cattle and cats, United States, 2024. *Emerg. Infect. Dis.* **30**, 1335–1343 (2024).
5. US Centers for Disease Control, "H5 bird flu: Current situation" (2025); www.cdc.gov/bird-flu/situation-summary/index.html.
6. S. Garg, K. Reinhart, A. Couture, K. Kniss, C. T. Davis, M. K. Kirby, E. L. Murray, S. Zhu, V. Kraushaar, D. A. Wadford, C. Drehoff, A. Kohnen, M. Owen, J. Morse, S. Eckel, J. Goswitz, G. Turabelidze, S. Krager, A. Unutzer, E. R. Gonzales, C. Abdul Hamid, S. Ellington, A. M. Mellis, A. Budd, J. R. Barnes, M. Biggerstaff, M. A. Jhung, M. Richmond-Crum, E. Burns, T. T. Shimabukuro, T. M. Uyeki, V. G. Dugan, C. Reed, S. J. Olsen, Highly pathogenic avian influenza A(H5N1) virus infections in humans. *N. Engl. J. Med.* **392**, 843–854 (2025).
7. US Department of Agriculture Animal and Plant Health Inspection Service, "Confirmations of highly pathogenic avian influenza in commercial and backyard flocks"; www.aphis.usda.gov/livestock-poultry-disease/avian/avian-influenza/hpai-detections/commercial-backyard-flocks.
8. T. P. Peacock, L. Moncla, G. Dudas, D. VanInsberghe, K. Sukhova, J. O. Lloyd-Smith, M. Worobey, A. C. Lowen, M. I. Nelson, The global H5N1 influenza panzootic in mammals. *Nature* **637**, 304–313 (2025).
9. R. Nataraj, A. Chandra, S. Kesavardhana, Avian influenza virus neuraminidase stalk length and haemagglutinin glycosylation patterns reveal molecularly directed reassortment promoting the emergence of highly pathogenic clade 2.3.4.4b A (H5N1) viruses. bioRxiv 595329 [Preprint] (2024); https://doi.org/10.1101/2024.05.22.595329.
10. M. R. Good, M. L. Fernández-Quintero, W. Ji, A. J. Rodriguez, J. Han, A. B. Ward, J. J. Guthmiller, A single mutation in dairy cow-associated H5N1 viruses increases receptor binding breadth. *Nat. Commun.* **15**, 10768 (2024).
11. B. Dadonaite, J. J. Ahn, J. T. Ort, J. Yu, C. Furey, A. Dosey, W. W. Hannon, A. L. Vincent Baker, R. J. Webby, N. P. King, Y. Liu, S. E. Hensley, T. P. Peacock, L. H. Moncla, J. D. Bloom, Deep mutational scanning of H5 hemagglutinin to inform influenza virus surveillance. *PLOS Biol.* **22**, e3002916 (2024).
12. T.-H. Lin, X. Zhu, S. Wang, D. Zhang, R. McBride, W. Yu, S. Babarinde, J. C. Paulson, I. A. Wilson, A single mutation in bovine influenza H5N1 hemagglutinin switches specificity to human receptors. *Science* **386**, 1128–1134 (2024).
13. D. W. Hawman, T. Tipih, E. Hodge, E. T. Stone, N. Warner, N. McCarthy, B. Granger, K. Meade-White, S. Leventhal, K. Hatzakis, S. Park, K. Gaffney, K. Rosenke, J. H. Erasmus, H. Feldmann, Clade 2.3.4.4b but not historical clade 1 HA replicating RNA vaccine protects against bovine H5N1 challenge in mice. *Nat. Commun.* **16**, 655 (2025).
14. M. J. Hogan, N. Pardi, MRNA vaccines in the COVID-19 pandemic and beyond. *Annu. Rev. Med.* **73**, 17–39 (2022).
15. C. Furey, G. Scher, N. Ye, L. Kercher, J. DeBeauchamp, J. C. Crumpton, T. Jeevan, C. Patton, J. Franks, A. Rubrum, M.-G. Alameh, S. H. Y. Fan, A. T. Phan, C. A. Hunter, R. J. Webby, D. Weissman, S. E. Hensley, Development of a nucleoside-modified mRNA vaccine against clade 2.3.4.4b H5 highly pathogenic avian influenza virus. *Nat. Commun.* **15**, 4350 (2024).
16. M. Hatta, Y. Hatta, A. Choi, J. Hossain, C. Feng, M. W. Keller, J. M. Ritter, Y. Huang, E. Fang, E. A. Pusch, T. Rowe, J. A. De La Cruz, M. C. Johnson, J. Liddell, N. Jiang, D. Stadlbauer, L. Liu, A. K. Bhattacharjee, J. R. Rouse, M. Currier, L. Wang, M. Z. Levine, M. K. Kirby, J. Steel, H. Di, J. R. Barnes, C. Henry, C. T. Davis, R. Nachbagauer, D. E. Wentworth, B. Zhou, An influenza mRNA vaccine protects ferrets from lethal infection with highly pathogenic avian influenza A(H5N1) virus. *Sci. Transl. Med.* **16**, eads1273 (2024).
17. X. Cui, P. Vervaeke, Y. Gao, L. Opsomer, Q. Sun, J. Snoeck, B. Devriendt, Z. Zhong, N. N. Sanders, Immunogenicity and biodistribution of lipid nanoparticle formulated self-amplifying mRNA vaccines against H5 avian influenza. *NPJ Vaccines* **9**, 138 (2024).
18. S. Chiba, M. Kiso, S. Yamada, K. Someya, Y. Onodera, A. Yamaguchi, S. Matsunaga, R. Uraki, K. Iwatsuki-Horimoto, S. Yamayoshi, F. Takeshita, Y. Kawaoka, An mRNA vaccine candidate encoding H5HA clade 2.3.4.4b protects mice from clade 2.3.2.1a virus infection. *NPJ Vaccines* **9**, 189 (2024).
19. C. P. Arevalo, M. J. Bolton, V. Le Sage, N. Ye, C. Furey, H. Muramatsu, M.-G. Alameh, N. Pardi, E. M. Drapeau, K. Parkhouse, T. Garretson, J. S. Morris, L. H. Moncla, Y. K. Tam, S. H. Y. Fan, S. S. Lakdawala, D. Weissman, S. E. Hensley, A multivalent nucleoside-modified mRNA vaccine against all known influenza virus subtypes. *Science* **378**, 899–904 (2022).
20. Arcturus Therapeutics Inc., "Safety and immunogenicity study of self-amplifying RNA pandemic influenza vaccine in adults" (ClinicalTrials.gov, 2026); https://clinicaltrials.gov/study/NCT06602531.
21. G. G. Hendricks, L. Grigoryan, M. J. Navarro, N. J. Catanzaro, M. L. Hubbard, J. M. Powers, M. Mattocks, C. Treichel, A. C. Walls, J. Lee, D. Ellis, J. Y. (J.) Wang, S. Cheng, M. C. Miranda, A. Valdez, C. W. Chao, S. Chan, C. Men, M. R. Johnson, H. Hui, S.-Y. Wu, V. Lujan, H. Muramatsu, P. J. C. Lin, M. M. H. Sung, Y. K. Tam, E. M. Leaf, N. Pardi, R. S. Baric, B. Pulendran, D. Veesler, A. Schäfer, N. P. King, Computationally designed mRNA-launched protein nanoparticle immunogens elicit protective antibody and T cell responses in mice. *Sci. Transl. Med.* **17**, eadu2085 (2025).
22. R. W. Sanders, J. P. Moore, Virus vaccines: Proteins prefer prolines. *Cell Host Microbe* **29**, 327–333 (2021).
23. M. Kishko, A. Stuebler, S. Sasmal, Y. Chan, D. Huang, C. Reyes, J. Lin, O. Price, A. Kume, K. Zong, C. Bricault, J. Alamares-Sapuay, L. Zhang, A computationally designed prefusion stabilized human metapneumovirus fusion protein vaccine antigen elicited a potent neutralization response. *Vaccine* **13**, 523 (2025).
24. A. Krarup, D. Truan, P. Furmanova-Hollenstein, L. Bogaert, P. Bouchier, I. J. M. Bisschop, M. N. Widjojoatmodjo, R. Zahn, H. Schuitemaker, J. S. McLellan, J. P. M. Langedijk, A highly stable prefusion RSV F vaccine derived from structural analysis of the fusion mechanism. *Nat. Commun.* **6**, 8143 (2015).
25. M. G. Joyce, B. Zhang, L. Ou, M. Chen, G.-Y. Chuang, A. Druz, W.-P. Kong, Y.-T. Lai, E. J. Rundlet, Y. Tsybovsky, Y. Yang, I. S. Georgiev, M. Guttman, C. R. Lees, M. Pancera, M. Sastry, C. Soto, G. B. E. Stewart-Jones, P. V. Thomas, J. G. Van Galen, U. Baxa, K. K. Lee, J. R. Mascola, B. S. Graham, P. D. Kwong, Iterative structure-based improvement of a fusion-glycoprotein vaccine against RSV. *Nat. Struct. Mol. Biol.* **23**, 811–820 (2016).
26. C.-L. Hsieh, S. A. Rush, C. Palomo, C.-W. Chou, W. Pickens, V. Más, J. S. McLellan, Structure-based design of prefusion-stabilized human metapneumovirus fusion proteins. *Nat. Commun.* **13**, 1299 (2022).
27. J. Gorman, C. S.-F. Cheung, Z. Duan, L. Ou, M. Wang, X. Chen, C. Cheng, A. Biju, Y. Sun, P. Wang, Y. Yang, B. Zhang, J. C. Boyington, T. Bylund, S. Charaf, S. J. Chen, H. Du, A. R. Henry, T. Liu, E. K. Sarfo, C. A. Schramm, C.-H. Shen, T. Stephens, I.-T. Teng, J.-P. Todd, Y. Tsybovsky, R. Verardi, D. Wang, S. Wang, Z. Wang, C.-Y. Zheng, T. Zhou, D. C. Douek, J. R. Mascola, D. D. Ho, M. Ho, P. D. Kwong, Cleavage-intermediate Lassa virus trimer elicits neutralizing responses, identifies neutralizing nanobodies, and reveals an apex-situated site-of-vulnerability. *Nat. Commun.* **15**, 285 (2024).
28. F. J. Milder, M. Jongeneelen, T. Ritschel, P. Bouchier, I. J. M. Bisschop, M. de Man, D. Veldman, L. Le, B. Kaufmann, M. J. G. Bakkers, J. Juraszek, B. Brandenburg, J. P. M. Langedijk, Universal stabilization of the influenza hemagglutinin by structure-based redesign of the pH switch regions. *Proc. Natl. Acad. Sci. U.S.A.* **119**, e2115379119 (2022).
29. A. Hanson, M. Imai, M. Hatta, R. McBride, H. Imai, A. Taft, G. Zhong, T. Watanabe, Y. Suzuki, G. Neumann, J. C. Paulson, Y. Kawaoka, Identification of stabilizing mutations in an H5 hemagglutinin influenza virus protein. *J. Virol.* **90**, 2981–2992 (2015).
30. S. E. Galloway, M. L. Reed, C. J. Russell, D. A. Steinhauer, Influenza HA subtypes demonstrate divergent phenotypes for cleavage activation and pH of fusion: Implications for host range and adaptation. *PLOS Pathog.* **9**, e1003151 (2013).
31. S.-S. Wong, J. DeBeauchamp, M. Zanin, Y. Sun, L. Tang, R. Webby, H5N1 influenza vaccine induces a less robust neutralizing antibody response than seasonal trivalent and H7N9 influenza vaccines. *NPJ Vaccines* **2**, 16 (2017).
32. H. M. Yassine, J. C. Boyington, P. M. McTamney, C.-J. Wei, M. Kanekiyo, W.-P. Kong, J. R. Gallagher, L. Wang, Y. Zhang, M. G. Joyce, D. Lingwood, S. M. Moin, H. Andersen, Y. Okuno, S. S. Rao, A. K. Harris, P. D. Kwong, J. R. Mascola, G. J. Nabel, B. S. Graham, Hemagglutinin-stem nanoparticles generate heterosubtypic influenza protection. *Nat. Med.* **21**, 1065–1070 (2015).
33. D. F. Burke, D. J. Smith, A recommended numbering scheme for influenza A HA subtypes. *PLOS ONE* **9**, e112302 (2014).
34. I. D. Lutz, S. Wang, C. Norn, A. Courbet, A. J. Borst, Y. T. Zhao, A. Dosey, L. Cao, J. Xu, E. M. Leaf, C. Treichel, P. Litvicov, Z. Li, A. D. Goodson, P. Rivera-Sánchez, A.-M. Bratovianu, M. Baek, N. P. King, H. Ruohola-Baker, D. Baker, Top-down design of protein architectures with reinforcement learning. *Science* **380**, 266–273 (2023).
35. J. R. R. Whittle, A. K. Wheatley, L. Wu, D. Lingwood, M. Kanekiyo, S. S. Ma, S. R. Narpala, H. M. Yassine, G. M. Frank, J. W. Yewdell, J. E. Ledgerwood, C.-J. Wei, A. B. McDermott,

B. S. Graham, R. A. Koup, G. J. Nabel, Flow cytometry reveals that H5N1 vaccination elicits cross-reactive stem-directed antibodies from multiple Ig heavy-chain lineages. *J. Virol.* **88**, 4047–4057 (2014).
36. J. Martín, S. A. Wharton, Y. P. Lin, D. K. Takemoto, J. J. Skehel, D. C. Wiley, D. A. Steinhauer, Studies of the binding properties of influenza hemagglutinin receptor-site mutants. *Virology* **241**, 101–111 (1998).
37. A. A. Abu-Shmais, G. Freeman, A. Creanga, M. J. Vukovich, T. Malla, G. E. Mantus, G. D. Shimberg, R. A. Gillespie, V. Guerra Canedo, B. Dadonaite, M. D. Rodgers, A. J. Chopde, E. Bardwil-Lugones, T. Bylund, A. R. Henry, J. Roberts-Torres, T. S. Johnston, S. Smith, E. S. Yang, C. Cheng, E. L. Walker, M. Ravichandran, I. J. Gordon, T. S. Dittakavi, D. S. Reed, T. C. Pierson, L. Dropulic, J. D. Bloom, Y. Tsybovsky, E. A. Boritz, D. C. Douek, T. Zhou, M. Kanekiyo, S. F. Andrews, Cross-neutralizing and potent human monoclonal antibodies against historical and emerging H5Nx influenza viruses. *Nat. Microbiol.* **10**, 2903–2918 (2025).
38. C. Dreyfus, N. S. Laursen, T. Kwaks, D. Zuijdgeest, R. Khayat, D. C. Ekiert, J. H. Lee, Z. Metlagel, M. V. Bujny, M. Jongeneelen, R. van der Vlugt, M. Lamrani, H. J. W. M. Korse, E. Geelen, Ö. Sahin, M. Sieuwerts, J. P. J. Brakenhoff, R. Vogels, O. T. W. Li, L. L. M. Poon, M. Peiris, W. Koudstaal, A. B. Ward, I. A. Wilson, J. Goudsmit, R. H. E. Friesen, Highly conserved protective epitopes on influenza B viruses. *Science* **337**, 1343–1348 (2012).
39. S. Bangaru, S. Lang, M. Schotsaert, H. A. Vanderven, X. Zhu, N. Kose, R. Bombardi, J. A. Finn, S. J. Kent, P. Gilchuk, I. Gilchuk, H. L. Turner, A. García-Sastre, S. Li, A. B. Ward, I. A. Wilson, J. E. Crowe Jr., A site of vulnerability on the influenza virus hemagglutinin head domain trimer interface. *Cell* **177**, 1136–1152.e18 (2019).
40. Y. Z. Tan, P. R. Baldwin, J. H. Davis, J. R. Williamson, C. S. Potter, B. Carragher, D. Lyumkis, Addressing preferred specimen orientation in single-particle cryo-EM through tilting. *Nat. Methods* **14**, 793–796 (2017).
41. G. Ueda, A. Antanasijevic, J. A. Fallas, W. Sheffler, J. Copps, D. Ellis, G. B. Hutchinson, A. Moyer, A. Yasmeen, Y. Tsybovsky, Y.-J. Park, M. J. Bick, B. Sankaran, R. A. Gillespie, P. J. Brouwer, P. H. Zwart, D. Veesler, M. Kanekiyo, B. S. Graham, R. W. Sanders, J. P. Moore, P. J. Klasse, A. B. Ward, N. P. King, D. Baker, Tailored design of protein nanoparticle scaffolds for multivalent presentation of viral glycoprotein antigens. *eLife* **9**, e57659 (2020).
42. S. Boyoglu-Barnum, D. Ellis, R. A. Gillespie, G. B. Hutchinson, Y.-J. Park, S. M. Moin, O. J. Acton, R. Ravichandran, M. Murphy, D. Pettie, N. Matheson, L. Carter, A. Creanga, M. J. Watson, S. Kephart, S. Ataca, J. R. Vaile, G. Ueda, M. C. Crank, L. Stewart, K. K. Lee, M. Guttman, D. Baker, J. R. Mascola, D. Veesler, B. S. Graham, N. P. King, M. Kanekiyo, Quadrivalent influenza nanoparticle vaccines induce broad protection. *Nature* **592**, 623–628 (2021).
43. F. Broecker, S. T. H. Liu, W. Sun, F. Krammer, V. Simon, P. Palese, Immunodominance of antigenic site B in the hemagglutinin of the current H3N2 influenza virus in humans and mice. *J. Virol.* **92**, e01100-18 (2018).
44. A. Dosey, D. Ellis, S. Boyoglu-Barnum, H. Syeda, M. Saunders, M. Watson, J. C. Kraft, M. N. Pham, M. Guttman, K. K. Lee, M. Kanekiyo, N. P. King, Combinatorial immune refocusing within the influenza hemagglutinin head elicits cross-neutralizing antibody responses. bioRxiv 541996 [Preprint] (2023); https://doi.org/10.1101/2023.05.23.541996.
45. I. Del Moral-Sánchez, R. A. Russell, E. E. Schermer, C. A. Cottrell, J. D. Allen, A. Torrents de la Peña, C. C. LaBranche, S. Kumar, M. Crispin, A. B. Ward, D. C. Montefiori, Q. J. Sattentau, K. Sliepen, R. W. Sanders, High thermostability improves neutralizing antibody responses induced by native-like HIV-1 envelope trimers. *NPJ Vaccines* **7**, 27 (2022).
46. Y. Feng, K. Tran, S. Bale, S. Kumar, J. Guenaga, R. Wilson, N. de Val, H. Arendt, J. DeStefano, A. B. Ward, R. T. Wyatt, Thermostability of well-ordered HIV spikes correlates with the elicitation of autologous tier 2 neutralizing antibodies. *PLOS Pathog.* **12**, e1005767 (2016).
47. A. Aung, A. Cui, L. Maiorino, A. P. Amini, J. R. Gregory, M. Bukenya, Y. Zhang, H. Lee, C. A. Cottrell, D. M. Morgan, M. Silva, H. Suh, J. D. Kirkpatrick, P. Amlashi, T. Remba, L. M. Froehle, S. Xiao, W. Abraham, J. Adams, J. C. Love, P. Huyett, D. S. Kwon, N. Hacohen, W. R. Schief, S. N. Bhatia, D. J. Irvine, Low protease activity in B cell follicles promotes retention of intact antigens after immunization. *Science* **379**, eabn8934 (2023).
48. D. Angeletti, J. S. Gibbs, M. Angel, I. Kosik, H. D. Hickman, G. M. Frank, S. R. Das, A. K. Wheatley, M. Prabhakaran, D. J. Leggat, A. B. McDermott, J. W. Yewdell, Defining B cell immunodominance to viruses. *Nat. Immunol.* **18**, 456–463 (2017).
49. K. M. Cirelli, D. G. Carnathan, B. Nogal, J. T. Martin, O. L. Rodriguez, A. A. Upadhyay, C. A. Enemuo, E. H. Gebru, Y. Choe, F. Viviano, C. Nakao, M. G. Pauthner, S. Reiss, C. A. Cottrell, M. L. Smith, R. Bastidas, W. Gibson, A. N. Wolabaugh, M. B. Melo, B. Cossette, V. Kumar, N. B. Patel, T. Tokatlian, S. Menis, D. W. Kulp, D. R. Burton, B. Murrell, W. R. Schief, S. E. Bosinger, A. B. Ward, C. T. Watson, G. Silvestri, D. J. Irvine, S. Crotty, Slow delivery immunization enhances HIV neutralizing antibody and germinal center responses via modulation of immunodominance. *Cell* **177**, 1153–1171.e28 (2019).
50. M. Kanekiyo, B. S. Graham, Next-generation influenza vaccines. *Cold Spring Harb. Perspect. Med.* **11**, a038448 (2021).
51. B. Shi, L. Ma, X. He, X. Wang, P. Wang, L. Zhou, X. Yao, Comparative analysis of human and mouse immunoglobulin variable heavy regions from IMGT/LIGM-DB with IMGT/HighV-QUEST. *Theor. Biol. Med. Model.* **11**, 30 (2014).
52. S. Cobey, S. E. Hensley, Immune history and influenza virus susceptibility. *Curr. Opin. Virol.* **22**, 105–111 (2017).
53. B. Dadonaite, J. Bloom, HA sequencing numbering (2024); https://dms-vep.org/Flu_H5_American-Wigeon_South-Carolina_2021-H5N1_DMS/numbering.html.
54. D. H. Barouch, Z.-Y. Yang, W.-P. Kong, B. Korioth-Schmitz, S. M. Sumida, D. M. Truitt, M. G. Kishko, J. C. Arthur, A. Miura, J. R. Mascola, N. L. Letvin, G. J. Nabel, A human T-cell leukemia virus type 1 regulatory element enhances the immunogenicity of human immunodeficiency virus type 1 DNA vaccines in mice and nonhuman primates. *J. Virol.* **79**, 8828–8834 (2005).
55. M. Kanekiyo, C.-J. Wei, H. M. Yassine, P. M. McTamney, J. C. Boyington, J. R. R. Whittle, S. S. Rao, W.-P. Kong, L. Wang, G. J. Nabel, Self-assembling influenza nanoparticle vaccines elicit broadly neutralizing H1N1 antibodies. *Nature* **499**, 102–106 (2013).
56. A. Punjani, J. L. Rubinstein, D. J. Fleet, M. A. Brubaker, cryoSPARC: Algorithms for rapid unsupervised cryo-EM structure determination. *Nat. Methods* **14**, 290–296 (2017).
57. M. Waskom, Seaborn: Statistical data visualization. *J. Open Source Softw.* **6**, 3021 (2021).
58. D. N. Mastronarde, SerialEM: A program for automated tilt series acquisition on Tecnai microscopes using prediction of specimen position. *Microsc. Microanal.* **9**, 1182–1183 (2003).
59. J. Abramson, J. Adler, J. Dunger, R. Evans, T. Green, A. Pritzel, O. Ronneberger, L. Willmore, A. J. Ballard, J. Bambrick, S. W. Bodenstein, D. A. Evans, C.-C. Hung, M. O'Neill, D. Reiman, K. Tunyasuvunakool, Z. Wu, A. Žemgulytė, E. Arvaniti, C. Beattie, O. Bertolli, A. Bridgland, A. Cherepanov, M. Congreve, A. I. Cowen-Rivers, A. Cowie, M. Figurnov, F. B. Fuchs, H. Gladman, R. Jain, Y. A. Khan, C. M. R. Low, K. Perlin, A. Potapenko, P. Savy, S. Singh, A. Stecula, A. Thillaisundaram, C. Tong, S. Yakneen, E. D. Zhong, M. Zielinski, A. Žídek, V. Bapst, P. Kohli, M. Jaderberg, D. Hassabis, J. M. Jumper, Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **630**, 493–500 (2024).
60. E. F. Pettersen, T. D. Goddard, C. C. Huang, G. S. Couch, D. M. Greenblatt, E. C. Meng, T. E. Ferrin, UCSF Chimera—A visualization system for exploratory research and analysis. *J. Comput. Chem.* **25**, 1605–1612 (2004).
61. C. M. Ives, O. Singh, S. D'Andrea, C. A. Fogarty, A. M. Harbison, A. Satheesan, B. Tropea, E. Fadda, Restoring protein glycosylation with GlycoShape. *Nat. Methods* **21**, 2117–2127 (2024).
62. T. I. Croll, ISOLDE: A physically realistic environment for model building into low-resolution electron-density maps. *Acta Crystallogr. D Struct. Biol.* **74**, 519–530 (2018).
63. D. Liebschner, P. V. Afonine, M. L. Baker, G. Bunkóczi, V. B. Chen, T. I. Croll, B. Hintze, L. W. Hung, S. Jain, A. J. McCoy, N. W. Moriarty, R. D. Oeffner, B. K. Poon, M. G. Prisant, R. J. Read, J. S. Richardson, D. C. Richardson, M. D. Sammito, O. V. Sobolev, D. H. Stockwell, T. C. Terwilliger, A. G. Urzhumtsev, L. L. Videau, C. J. Williams, P. D. Adams, Macromolecular structure determination using X-rays, neutrons and electrons: Recent developments in Phenix. *Acta Crystallogr. D Struct. Biol.* **75**, 861–877 (2019).
64. X. Du, S. M. Ansell, "Lipids and lipid nanoparticle formulations for delivery of nucleic acids," world patent WO/2017/004143 (2017); https://patentscope.wipo.int/search/en/WO2017004143.
65. A. Creanga, R. A. Gillespie, B. E. Fisher, S. F. Andrews, J. Lederhofer, C. Yap, L. Hatch, T. Stephens, Y. Tsybovsky, M. C. Crank, J. E. Ledgerwood, A. B. McDermott, J. R. Mascola, B. S. Graham, M. Kanekiyo, A comprehensive influenza reporter virus panel for high-throughput deep profiling of neutralizing antibodies. *Nat. Commun.* **12**, 1722 (2021).
66. B. Dadonaite, K. H. D. Crawford, C. E. Radford, A. G. Farrell, T. C. Yu, W. W. Hannon, P. Zhou, R. Andrabi, D. R. Burton, L. Liu, D. D. Ho, H. Y. Chu, R. A. Neher, J. D. Bloom, A pseudovirus system enables deep mutational scanning of the full SARS-CoV-2 spike. *Cell* **186**, 1263–1278.e20 (2023).
67. B. Dadonaite, J. Bloom, Escape from vaccine-elicited serum antibody neutralization (2025); https://dms-vep.org/Flu_H5_American-Wigeon_South-Carolina_2021-H5N1_DMS/htmls/Stabilized_HA_sera_escape_faceted.html.
68. B. Dadonaite, J. Bloom, W. Hannon, jennyahn, dms-vep/Flu_H5_American-Wigeon_South-Carolina_2021-H5N1_DMS: VRC_mAb_publication, Zenodo (2025); http://dx.doi.org/10.5281/ZENODO.16740862.
69. B. Dadonaite, J. Bloom, W. Hannon, J. Ahn, Flu_H5_American-Wigeon_South-Carolina_2021-H5N1_DMS: Deep mutational scanning of H5 VEP, Github (2025); https://github.com/dms-vep/Flu_H5_American-Wigeon_South-Carolina_2021-H5N1_DMS.

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AI141707 (to J.D.B.), National Institute of Allergy and Infectious Diseases grant 75N93021C00015 (to J.D.B.), and the intramural research program of the Vaccine Research Center, National Institute of Allergy and Infectious Diseases, National Institutes of Health (to M.K.). **Author contributions:** A.D., M.K., and N.P.K. designed the study. A.D. designed stabilizing mutations and produced proteins. A.D. and M.J.V. performed BLI, and A.D. did all other in vitro characterization. A.D. carried out cryo-EM and model building. E.M.L., J.M., and E.G. immunized mice for the immunization-only study. H.M., R.H.J.J., and N.P. made the mRNA-LNPs. A.D. carried out ELISAs, HAI, and EMPEM. R.A.G. carried out neutralization assays. R.A.G. and M.J.K. carried out all immunizations and assays related to immunization and challenge study. B.D. and J.D.B. carried out DMS experiments. A.D., B.D., and N.P.K. wrote the manuscript with input from all authors. **Competing interests:** A.D., M.K., and N.P.K. are inventors on University of Washington licensed patents related to influenza vaccines (WO2023133435A9, titled "Novel immunogens for influenza virus vaccines"). N.P.K. consults for AstraZeneca. J.D.B. and B.D. are inventors on Fred Hutchinson Cancer Center–licensed patents related to deep mutational scanning of viral proteins (WO2024081810A2, titled "Pseudo-viral systems for mutational scanning of viral proteins"). J.D.B. consults for Apriori Bio, Invivyd, GSK, Pfizer, and the Vaccine Company. N.P. is named on patents describing the use of nucleoside-modified mRNA in lipid nanoparticles as a vaccine platform (US20190274968A1, titled "Nucleoside-modified RNA for inducing an adaptive immune response"). J.D.B. has disclosed those interests fully to the University of Pennsylvania and has in place an approved plan for managing any potential conflicts arising from licensing of those patents. N.P. served on the mRNA strategic advisory board of Sanofi Pasteur in 2022, Pfizer in 2023–2024, and AldexChem in 2023 to 2025. N.P. is a member of the scientific advisory board of BioNet and Piezo Therapeutics and has consulted for Vaccine Company Inc. and Pasture Bio. R.H.J.J. is an employee of Acuitas Therapeutics, a company developing LNPs for the delivery of mRNA-based therapeutics, and this company is on a related patent (WO 2017/004143A1, titled "Lipids and lipid nanoparticle formulations for the delivery of nucleic acids"). The other authors declare that they have no competing interests. **Data, code, and materials availability:** All data associated with this study are present in the paper or the Supplementary Materials. All images and data were generated and analyzed by the authors. nsEM maps have been deposited at the Electron Microscopy Data Bank (EMDB) with accession codes EMD-71459 (Fab 3 reconstruction of polyclonal serum from mouse immunized with H5 TX24-FMLMI-RC_I_1 in complex with TX24-I53_dn5B), EMD-71460 (Fab 2 reconstruction of polyclonal serum from mouse immunized with H5 TX24-FMLMI-RC_I_1 in complex with TX24-I53_dn5B), EMD-71461 (Fab 1 reconstruction of polyclonal serum from mouse immunized with H5 TX24-FMLMI-RC_I_1 in complex with TX24-I53_dn5B), EMD-71462 (Fab reconstruction of polyclonal serum from mouse immunized with H5 TX24-foldon in complex with TX24-I53_dn5B), EMD-71463 (Fab 1 reconstruction of polyclonal serum from mouse immunized with H5 TX24-FMLMI-foldon in complex with TX24-I53_dn5B), EMD-71464 (Fab 2 reconstruction of polyclonal serum from mouse immunized with H5 TX24-FMLMI-foldon in complex with TX24-I53_dn5B), EMD-71465 (Fab 3 reconstruction of polyclonal serum from mouse immunized with H5 TX24-FMLMI-foldon in complex with TX24-I53_dn5B), EMD-71466 (Fab 1 reconstruction of polyclonal serum from mouse immunized with the H5 TX24–membrane-anchored construct in complex with TX24-I53_dn5B), EMD-71467 (Fab 2 reconstruction of polyclonal serum from mouse immunized with the H5 TX24–membrane-anchored construct in complex with TX24-I53_dn5B), EMD-71468 (Fab 3 reconstruction of polyclonal serum from mouse immunized with the H5 TX24–membrane-anchored construct in complex with TX24-I53_dn5B), EMD-71469 (Fab 4 reconstruction of polyclonal serum from mouse immunized with the H5 TX24–membrane-anchored construct in complex with the TX24-I53_dn5B), EMD-71470 (Fab 1 reconstruction of polyclonal serum from mouse immunized with the H5 TX24-FMLMI–membrane-anchored construct in complex with TX24-I53_dn5B), EMD-71471 (Fab 2 reconstruction of polyclonal serum from mouse immunized with the H5 TX24-FMLMI–membrane-anchored construct in complex with TX24-I53_dn5B), EMD-71472 (Fab 3 reconstruction of polyclonal serum from mouse immunized with the H5 TX24-FMLMI–membrane-anchored construct in complex with TX24-I53_dn5B), and EMD-71473 (Fab 4 reconstruction of polyclonal serum from mouse immunized with the H5 TX24-FMLMI–membrane-anchored construct in complex with TX24-I53_dn5B). The cryo-EM map and model have been deposited at the EMDB with accession code EMD-71796 and at the Research Collaboratory for Structural Bioinformatics Protein Data Bank (RCSB PDB) with accession code 9PR3. All vaccine constructs will be made available on request after completion of a material transfer agreement. All other materials used or generated in this study are commercially available or will be supplied upon reasonable request.

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Stabilization of the H5 clade 2.3.4.4b hemagglutinin improves vaccine-elicited neutralizing antibody responses in mice

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Editor's summary
The concept of a stabilized vaccine immunogen hit prime time during the COVID-19 pandemic, where the SARS-CoV-2 spike protein was stabilized to elicit optimal immunity. Other vaccines, such as those against respiratory syncytial virus (RSV), have used similar tactics. Here, Dosey *et al.* asked whether stabilization could improve the immunogenicity of the influenza virus H5 hemagglutinin, which is less stable than its seasonal counterparts. The authors generated mRNA- and protein-based vaccines comparing stabilized H5 HA with an unstabilized, wild-type version and demonstrated consistent superiority of the stabilized immunogen. Stabilization not only quantitatively changed the antibody response; the authors also observed a shift in antibody binding location toward the receptor binding site of the HA. The authors lastly demonstrated that the stabilized H5 HA vaccine could confer protection against H5N1 challenge in mice, providing support for moving this stabilized vaccine immunogen forward. —Courtney Malo

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